Spitfire

Ti:Sapphire Regenerative Amplifier Systems

Spitfire F
Spitfire USF
Spitfire 50FS
Spitfire P
Spitfire PM

User's Manual

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Preface

This manual contains information you need to safely operate and maintain your Spectra-Physics *Spitfire* Ti:sapphire amplifier system. The *Spitfire* is available in a wide variety of models; this manual covers the *Spitfire F*, *P*, *PM*, *USF*, and *50FS* versions. Other versions of the *Spitfire*, such as the *Spitfire HP*, are described in their own manuals.

The *Spitfire* systems amplify short duration optical pulses emitted by mode-locked, Ti:sapphire lasers, such as those produced by the Spectra-Physics *Tsunami* or *Mai Tai*. The *Spitfire* can amplify either picosecond pulses or femtosecond pulses at near infrared and red wavelengths. Two basic repetition rates are available, 1 kHz and 5 kHz, and the system can be adjusted for lower pulse repetition rates.

The system comprises two units: the *Spitfire* head assembly and its control unit, the Synchronous Delay Generator, or *SDG II*. The *SDG II* is a table-top unit that is provided with all systems. The *Spitfire* amplifier head itself contains three assemblies: a pulse stretcher, a Ti:sapphire regenerative amplifier and a pulse compressor.

The *Spitfire* stretcher and compressor designs are based on the pulse width of the input and output pulses. The optics, including the pulse stretcher and compressor, are optimized for the range of wavelength, pulse width, and repetition rate used. This manual contains information on the optics sets and the stretcher and compressor configurations available for this system.

Please note that the *Spitfire* performance specifications can be met only if the mode-locked Ti:sapphire laser is operating within the specifications and requirements outlined in this manual. The amplifier is designed specifically for the Spectra-Physics *Tsunami* or *Mai Tai* lasers.

The “Introduction” contains a brief description of the *Spitfire* head assembly and the *SDG II* controller.

Following that section is an important chapter on laser safety. The *Spitfire* is a Class IV laser and, as such, emits laser radiation which can permanently damage eyes and skin. This section contains information about these hazards and offers suggestions on how to safeguard against them.

“General Description,” contains an introductory section on laser theory, pulse stretching, laser amplification and pulse compression. Specifications for the various *Spitfire* systems are included at the end of this chapter.

The next chapter is an overview of the external controls and external adjustments of the system. Please familiarize yourself with this material before operating the amplifier.

The following chapter describes the preparation needed to install the *Spitfire* system. While this manual contains a brief installation procedure, it is only a guide to preparing the site for the initial set-up of the *Spitfire* system. Please wait for the Spectra-Physics service engineer to install the system as part of your purchase agreement. Only personnel authorized by Spectra-Physics can install and set up your *Spitfire* system.

"Operation" describes the routine operation of the *Spitfire*. It is followed by a detailed description of the beam path and internal adjustments of the amplifier. This information is needed should it become necessary to perform a simple re-alignment of the system using the procedures described in this manual. A full alignment of the system should only be performed by an authorized Spectra-Physics representative.

The "Maintenance and Troubleshooting" section contains maintenance procedures that will allow you to keep your *Spitfire* clean and operational on a day-to-day basis. It also contains procedures you can perform to remedy any minor problems that might be encountered. Also included are procedures to help you guide your Spectra-Physics field service engineer to the source of any major problems. *Do not attempt repairs yourself while the unit is still under warranty*; instead, report all problems to Spectra-Physics for warranty repair. This section includes a replacement parts list plus a list of world-wide Spectra-Physics service centers you can call if you need help.

This product has been tested and found to conform to "Directive 89/336/EEC for Electromagnetic Compatibility." Class A compliance was demonstrated for "EN 50081-2:1993 Emissions" and "EN 50082-1:1992 Immunity" as listed in the official Journal of the European Communities. Refer to "CE Declaration of Conformity (Low Emissions)" on page 2-7.

Every effort has been made to ensure that the information in this manual is accurate. All information in this document is subject to change without notice. Spectra-Physics makes no representation or warranty, either express or implied with respect to this document. In no event will Spectra-Physics be liable for any direct, indirect, special, incidental or consequential damages resulting from any defects in this documentation. If you encounter any difficulty with the content or style of this manual, please let us know. The last page is a form to aid in bringing such problems to our attention.

Thank you for your purchase of Spectra-Physics instruments.

Environmental Specifications

CE Electrical Equipment Requirements

For information regarding the equipment needed to provide the electrical service listed under "Required Utilities" on page 5-3, please refer to specification EN-309, "Plug, Outlet and Socket Couplers for Industrial Uses," listed in the official *Journal of the European Communities.*

Environmental Specifications

The environmental conditions under which the laser system will function are listed below:

For indoor use only.

Altitude:	up to 2000 m
Temperatures:	10° C to 40° C
Maximum relative humidity:	80% non-condensing for temperatures up to 31° C.
Mains supply voltage:	do not exceed ± 10% of the nominal voltage
Insulation category:	II
Pollution degree:	2

FCC Regulations

This equipment has been tested and found to comply with the limits for a Class A digital device pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with this instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense.

Modifications to the laser system not expressly approved by Spectra-Physics could void your right to operate the equipment.

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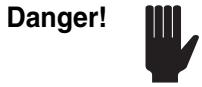
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Warning Conventions

The following warnings are used throughout this manual to draw your attention to situations or procedures that require extra attention. They warn of hazards to your health, damage to equipment, sensitive procedures, and exceptional circumstances. All messages are set apart by a thin line above and below the text as shown here.



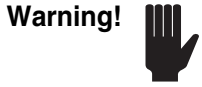
Laser radiation is present.



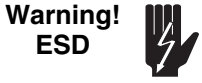
Condition or action may present a hazard to personal safety.



Condition or action may present an electrical hazard to personal safety.



Condition or action may cause damage to equipment.



Action may cause electrostatic discharge and cause damage to equipment.



Condition or action may cause poor performance or error.



Text describes exceptional circumstances or makes a special reference.



Do not touch.



Appropriate laser safety eyewear should be worn during this operation.



Refer to the manual before operating or using this device.

Standard Units

The following units, abbreviations, and prefixes are used in this Spectra-Physics manual:

Quantity	Unit	Abbreviation
mass	kilogram	kg
length	meter	m
time	second	s
frequency	hertz	Hz
force	newton	N
energy	joule	J
power	watt	W
electric current	ampere	A
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
inductance	henry	H
magnetic flux	weber	Wb
magnetic flux density	tesla	T
luminous intensity	candela	cd
temperature	celcius	C
pressure	pascal	Pa
capacitance	farad	F
angle	radian	rad

Prefixes								
tera	(10^{12})	T	deci	(10^{-1})	d	nano	(10^{-9})	n
giga	(10^{9})	G	centi	(10^{-2})	c	pico	(10^{-12})	p
mega	(10^{6})	M	mill	(10^{-3})	m	femto	(10^{-15})	f
kilo	(10^{3})	k	micro	(10^{-6})	μ	atto	(10^{-18})	a

Appreviations

The following is a list of abbreviations used in Spectra-Physics manuals:

ac	alternating current
AOM	acousto-optic modulator
APM	active pulse mode locking
AR	anti reflection
BI-FI	birefringent filter
CDRH	Center of Devices and Radiological Health
CE	European Union
CPM	colliding pulse mode locking
CW	continuous wave
dc	direct current
E/O	electro-optic
fs	femtosecond or 10^{-15} second
GTI	Gires-Toutnois Interferometer
GVD	group velocity dispersion
HR	high reflector
IR	infrared
OC	output coupler
PS	picosecond or 10^{-12} second
PZT	piezo-electric transducer
RF	radio frequency
SBR	saturable Bragg reflector
SCFH	standard cubic feet per hour
SPM	self phase modulation
TEM	transverse electromagnetic mode
TI:SAPPHIRE	Titanium-doped Sapphire
UV	ultraviolet
λ	wavelength

Unpacking and Inspection

Unpacking Your System

Your *Spitfire* laser amplifier was packed with great care, and the containers were inspected prior to shipment. Upon receiving the system, immediately inspect the outside of the shipping containers. If there is any major damage (holes in the containers, crushing, etc.), insist that a representative of the carrier be present when you unpack the contents.

Instructions for unpacking the system are attached to the outside of the containers. It is important that these instructions are followed carefully. The system was precisely aligned at the factory, then packed and shipped in a manner to preserve that alignment. Handle the system with care while unpacking to preserve this condition.

Carefully inspect the laser system as you unpack it. If any damage is evident, such as dents or scratches on the covers or broken parts, etc., immediately notify the carrier and your Spectra-Physics sales representative.

Keep the shipping containers. If you file a damage claim, they may be needed to demonstrate that the damage occurred as a result of shipping. If the system is ever returned for service, the specially designed containers assure adequate protection.

Warning!



Spectra-Physics considers itself responsible for the safety, reliability and performance of the *Spitfire* amplifier only under the following conditions:

- All field installable options, modifications or repairs are performed by persons trained and authorized by Spectra-Physics.
- The equipment is used in accordance with the instructions provided in this manual.

System Components

The system is shipped in two separate containers:

- One contains the *Spitfire* assembly
- One contains the *SDG II* controller and accessory kit (see below)

Accessory Kit

Included with the laser system is this manual, a packing slip listing all the components shipped with this order, and an accessory kit containing the following items:

- cables (kit)
- *SDG II* controller
- DC motor controller and AC adapter
- (4) chassis clamps
- beam tubes for the pump laser
- (2) routing mirror assemblies for the beam from the seed laser

Chapter 1 Introduction

The Spectra-Physics *Spitfire* system amplifies individual laser pulses that are selected from a stream of pulses and produced by a separate, mode-locked Ti:sapphire laser. Typically, an input pulse with an energy of only a few nanojoules can be amplified to about 1 millijoule. Specific *Spitfire* models can amplify pulses ranging in duration from less than 50 femtoseconds up to about 80 picoseconds.

The maximum output energy of a solid-state amplifier is normally limited by the optical damage threshold of the crystalline material used in the system. The *Spitfire* regenerative amplifier circumvents this limitation by using "chirped pulse amplification." This technique, originally developed for radar systems, first temporally stretches a pulse to reduce its peak power, then amplifies it, and finally recompresses the pulse to a width close to its original duration. This results in greatly increased peak power while avoiding optical damage to the amplifier.

The Spitfire System

The *Spitfire* system itself comprises two main components:

- the *Spitfire* amplifier head assembly, and the
- the Synchronization and Delay Generator (*SDG II*)

However, a complete system requires a pump laser to energize the *Spitfire* amplifier and a seed laser to provide the original pulses. Figure 1-1 shows a typical application: a *Spitfire PM* pumping an optical parametric amplifier, seeded by a *Mai Tai* laser system and pumped by an *Evolution* laser.

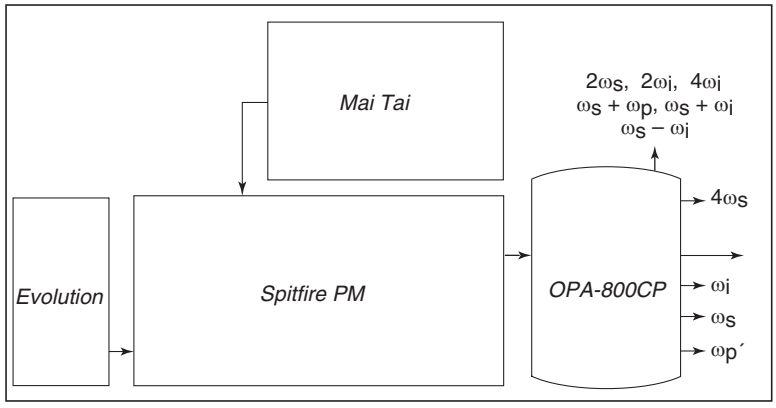


Figure 1-1: A typical layout showing the *Spitfire* pumping a Spectra-Physics *OPA-800CP.*

Configurations

The *Spitfire* system is available in a variety of models that produce a wide range of pulse durations. Table 1-1 lists the models covered by this manual. Each *Spitfire* model operates at either a 1 kHz or 5 kHz pulse repetition rate and can be ordered preset to either rate. Systems may also be converted from one models to another. Contact your Spectra-Physics representative for more information about these options.

Most of the configurations listed in Table 1-1 are also available with a high-power option, the *Spitfire HP*, that operates at 1 kHz. *Spitfire HP* systems add extra length to the head assembly to accommodate a second stage of amplification. These high-power systems are described in separate documentation. Contact your Spectra-Physics representative for more information.

Table 1-1: *Spitfire* Configuration Matrix

Amplifier Model	Pulse Width	Description
Spitfire F	<130 fs	“standard” version
Spitfire P	<2 ps	produces picosecond pulses using a picosecond seed laser
Spitfire PM	<2 ps	“pico-mask” version - produces picosecond pulses using a femtosecond seed laser
Spitfire USF	<90 fs	simple reconfiguration for shorter pulses
Spitfire 50FS	< 50 fs	ultra-short output pulses

All versions of the *Spitfire* listed in Table 1-1, with the exception of the *Spitfire 50FS*, are available in three standard wavelength ranges, as determined by the optics set used. The *Spitfire 50FS* is available only with Optics Set 1.

Table 1-2: *Spitfire* Optics Sets

Optics Set	Output Wavelength Range
Optics Set 1	750 nm – 840 nm
Optics Set 2	840 nm – 870 nm
Optics Set 3	870 nm – 900 nm

The wavelength range of interest was specified when your *Spitfire* system was ordered. But there are separate optics sets, depending on whether the system will be run at 1 kHz or 5 kHz.

Custom Spitfire Systems

Custom versions of the *Spitfire* are available that produce pulses at different wavelengths, higher power pulses or pulses at repetition rates other than those listed in Table 1-1. Again, contact your Spectra-Physics representative for more information.

The Spitfire Amplifier

The *Spitfire* amplifier contains the optics and opto-mechanical devices for stretching, selecting, amplifying and compressing pulses from a seed laser (such as a Spectra-Physics *Mai Tai* or *Tsunami*). The *Spitfire* amplifier comprises the following three assemblies:

- the optical pulse stretcher
- the regenerative amplifier
- the optical pulse compressor

These assemblies are each carefully optimized for the chosen wavelength range, the repetition rate of the amplified output, and the duration of the amplified pulses

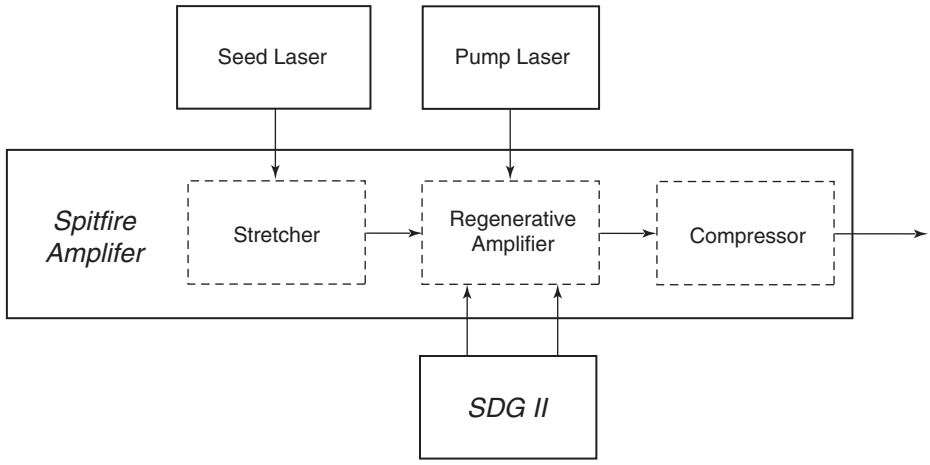


Figure 1-2: Block Diagram for the *Spitfire F, P, PM, USF* and *50FS*

Incoming seed pulses are stretched using a multi-pass grating and mirror combination. The *SDG II* provides the synchronization and control needed to select and capture individual pulses from the train of stretched seed pulses and direct them into the amplifier. The selected, stretched pulses then pass multiple times through the regenerative amplifier.

Once the pulses are amplified, the *SDG II* provides the timing control to direct the amplified pulses into the compressor. The compressor shortens the amplified pulses close to their original duration using a second grating/mirror combination. The pulses are then directed out of the *Spitfire*.

Titanium Sapphire

The *Spitfire* amplifier gain media is a titanium-doped sapphire (Ti:sapphire) crystal. Ti:sapphire was selected because of its two very useful properties: (a) it has a broad absorption band in the blue and green, which allows it to be pumped by the frequency-doubled output of a Nd:YLF or a Nd:YAG laser, and (b) it is tunable over a broad emission band of wavelengths in the near infrared. For a more detailed explanation of the theory of operation of the *Spitfire*, refer to Chapter 3, “General Description.”

Chapter 2 Laser Safety

Danger!





The Spectra-Physics *Spitfire*® amplifier is classified as a C*lass IV—High Power Laser* whose beam is, by definition, a safety and fire hazard. Take precautions to prevent accidental exposure to both direct and reflected beams. Diffuse as well as specular beam reflections can cause severe eye or skin damage.

Because the output wavelength is typically between 700 and 1000 nm (from red to infrared), the *Spitfire* output beam is often invisible and therefore especially dangerous. This type of infrared radiation passes easily through the cornea of the eye, and, when focused on the retina, can cause instantaneous and permanent damage.

Precautions For The Safe Operation Of Class IV High Power Lasers

- Wear protective eyewear at all times; selection depends on the wavelength and intensity of the radiation, the conditions of use, and the visual function required. Protective eyewear is available from suppliers listed in the *Laser Focus World*, *Lasers and Optronics*, and *Photonics Spectra* buyer's guides. Consult the ANSI and ACGIH standards listed at the end of this section for guidance.
- Maintain a high ambient light level in the laser operation area so the eye's pupil remains constricted, reducing the possibility of damage.
- Avoid looking at the output beam; even diffuse reflections are hazardous.
- Avoid blocking the output beam or its reflections with any part of the body.
- Establish a controlled access area for laser operation. Limit access to those trained in the principles of laser safety.
- Enclose beam paths wherever possible.
- Post prominent warning signs near the laser operating area (Figure 2-1).
- Set up experiments so the laser beam is either above or below eye level.
- Set up shields to prevent any unnecessary specular reflections or beams from escaping the laser operation area.
- Set up a beam dump to capture the laser beam and prevent accidental exposure (Figure 2-2).

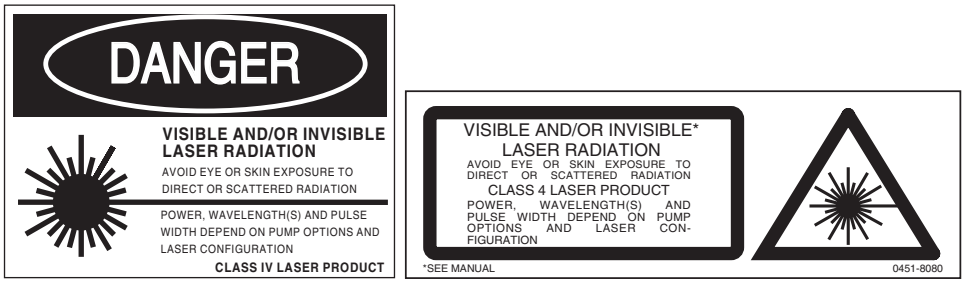


Figure 2-1: These CE and CDRH standard safety warning labels would be appropriate for use as entry warning signs (EN 60825.1, ANSI Z136.1 Section 4.7).

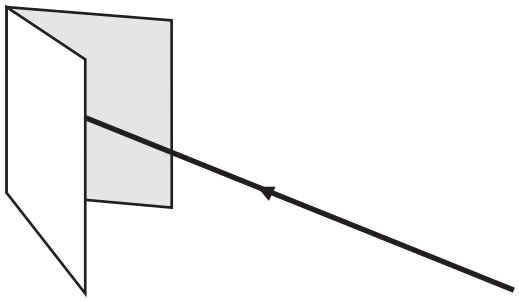


Figure 2-2: Folded Metal Beam Target

Caution! 

Use of controls or adjustments, or the performance of procedures other than those specified herein may result in hazardous radiation exposure.

Follow the instructions contained in this manual for safe operation of your laser. At all times during operation, maintenance, or service of your laser, avoid unnecessary exposure to laser or collateral radiation* that exceeds the accessible emission limits listed in "Performance Standards for Laser Products," *United States Code of Federal Regulations*, 21CFR1040 10(d).

Safety Devices

Because the *Spitfire* cannot generate output energy without being pumped and seeded by other lasers, it requires no safety interlocks or emission indicator. All safety interlocks and emission indicators are associated with the pump and seed lasers. When both the pump and seed lasers are disabled, the *Spitfire* is disabled.

* *Any electronic product radiation, except laser radiation, emitted by a laser product as a result of, or necessary for, the operation of a laser incorporated into that product.*

Fuses

The *Spitfire SDG II* controller uses one of the following fuses, as appropriate for the local line voltage:

120 Vac	220 Vac
F1AH 250 V, Slow Blow	F0.5AH 250 V, Slow Blow

Maximum Emission Levels

The following is the maximum emission level possible for the *Spitfire* amplifier. Use this information for selecting appropriate laser safety eyewear and implementing appropriate safety procedures. This value does not imply actual system power or specifications.

Emission Wavelength	Maximum Power
690 to 1080 nm	10 W

CDRH Compliance

This laser product complies with Title 21 of the *United States Code of Federal Regulations*, Chapter 1, subchapter J, parts 1040.10 and 1040.11, as applicable. To maintain compliance with these regulations, once a year, or whenever the product has been subjected to adverse environmental conditions (e.g., fire, flood, mechanical shock, spilled solvent, etc.), check to see that all features of the product identified on the CDRH Radiation Control Drawing (found later in this chapter) function properly. Also, make sure that all warning labels remain firmly attached.

CDRH Requirements for Operating the Spitfire Using the Optional PC Control

The *Spitfire* system complies with all CDRH safety standards when operated using the *SDG II* controller. However when the laser is operated from a computer using the command language described in Appendix A, "RS-232 Interface," the following must be provided in order to satisfy CDRH regulation requirements:

- **An emission indicator**—that indicates laser energy is present or can be accessed. It can be a "power-on" lamp, a computer display that flashes a statement to this effect, or an indicator on the control equipment for this purpose. It need not be marked as an emission indicator so long as its function is obvious. *Its presence is required on any control panel that affects laser output, including a computer display panel.*

CE/CDRH Radiation Control Drawings

Refer to the warning labels in Figure 2-4.

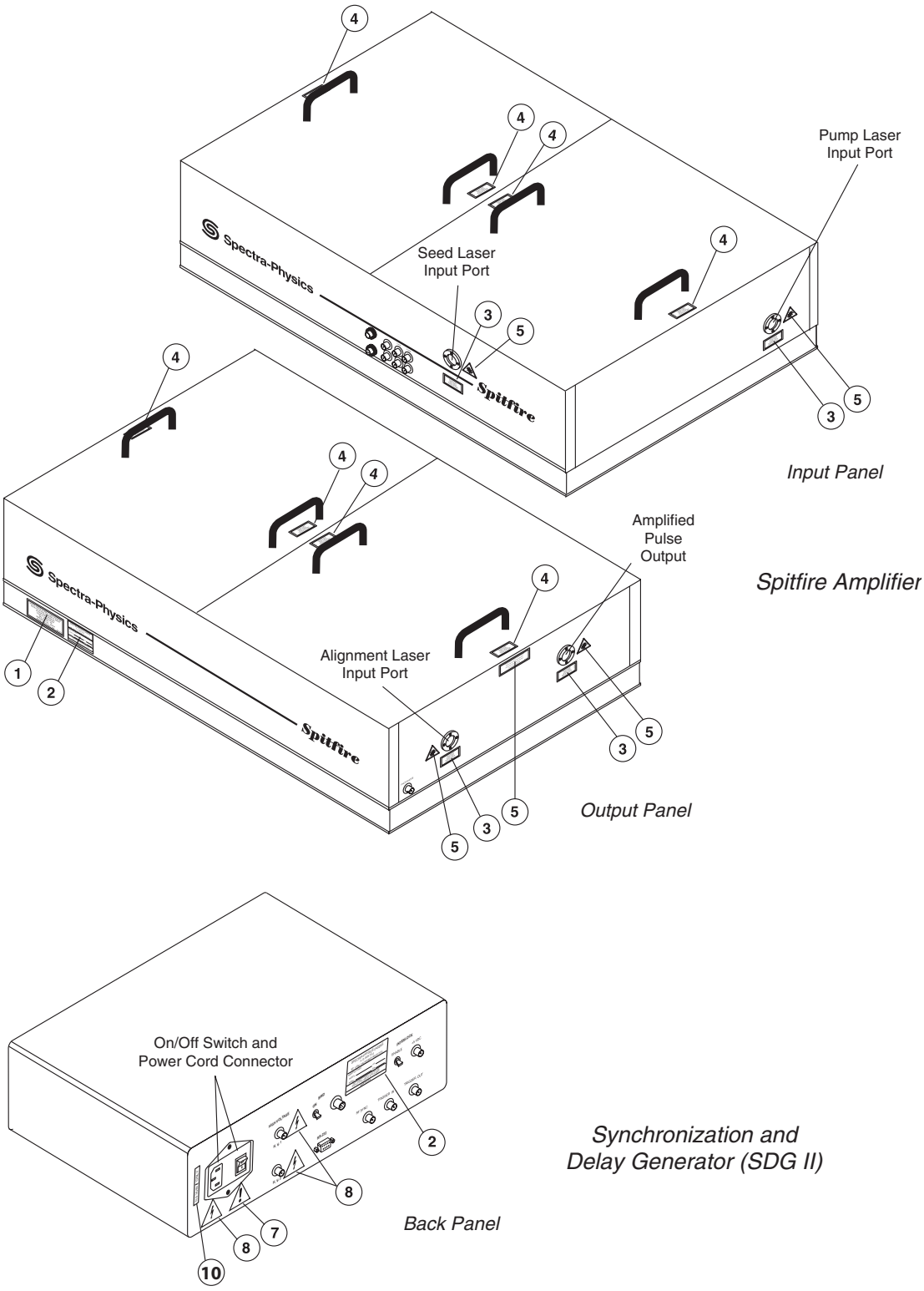
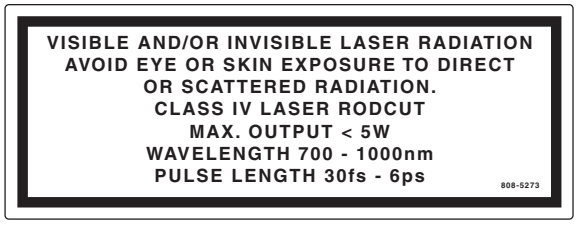
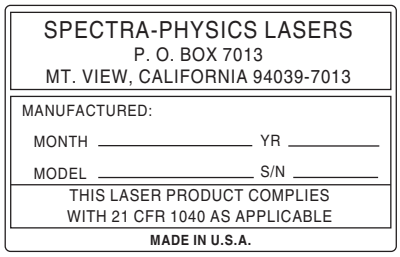


Figure 2-3: CE/CDRH Radiation Control Drawing

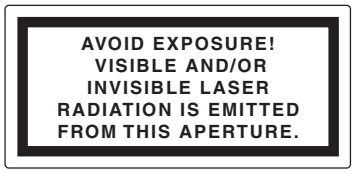
CE/CDRH Warning Labels



CE Warning Label (1)



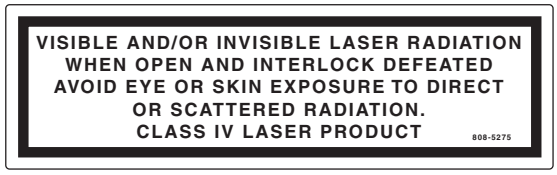
Identification/Certification Label (2)



CE Aperture Label (3)
Part 1



Caution Label
RF Energy Present (4)



Danger–Interlocked Housing Label (5)



CE Aperture Label (6)
Part 2



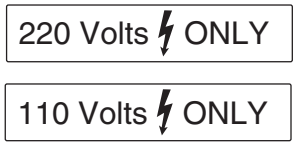
CE Caution Label (7)



CE Electrical Warning Label (8)



CE Certification Label (9)



Voltage Input Label (10)

Figure 2-4: CE/CDRH Warning Labels

Label Translations

For safety, the following translations are provided for non-English speaking personnel. The number in parenthesis in the first column corresponds to the label number listed on the previous page.

Table 2-1: Label Translations

Label #	French	German	Spanish	Dutch
CE Warning Label (1)	Rayonnement visible et/ou invisible exposition dangereuse de l'œil ou de la peau au rayonnement direct ou diffus. Appareil a laser de Classe 4. Puissance maximum 5 W. Longueur D'onde 700−1000 nm. Duree d'impulsion 30 fs−6 ps	Austritt von sichtbarer und/oder unsichtbarer Laserstrahlung. Augen- und Hautkontakt mit direkter Strahlung oder Streustrahlung vermeiden. Laser Klasse IV Maximale Ausgangsleistung < 5 W Wellenlänge 700 - 1000 nm Pulsbreite 30 fs - 6 ps	Radiación láser visible y/o invisible. Evitar la exposición de los ojos o la piel a la radiacion, ya sea directa ó difusa. Producto láser Clase IV. Potencia máxima <5 W. Longitud de onda: 700−1000 nm. Longitud de pulso: 30 fs−6 ps.	Zichtbare en/of onzichtbare* laser straling. Vermijd blootstelling aan ogen of huid door directe of gereflecteerde straling. Klasse 4 laser produkt; 532 nm, maximaal uittredend vermogen 15 W. *zie handleiding
CE Aperture Label (3)	Exposition Dangereuse – Un Rayonnement laser visible et/ou invisible est emis par cette ouverture.	Nicht dem Strahl aussetzen! Austritt von sichtbarer und/oder unsicht-barer Laserstrahlung.	! Evitar exponerse ¡ Atraves de esta apertura se emite radiacion laser visible y/o invisible.	Vanuit dit apertuur wordt zichtbare en onzichtbare laserstraling geemiteerd! Vermijd blootstelling!
CE Interlocked Label (4)	Rayonnement Laser Visible et/ou Invisible en Cas D'Ouverture et lorsque la securité est neutralisée; exposition dangereuse de l'oeil ou de la peau au rayonnement direct ou diffus. Laser de Classe 4.	Sichtbare und/oder unsichtbare Laserstrahlung wenn geöffnet und Sicherheitsverriegelung überbruckt. Bestrahlung von Augen oder Haut durch direkt oder Streustrahlung vermeiden. Laser Klasse 4.	Al abrir y retirar el dispositivo de seguridad exist radiacion laser visible y invisible; evite que los ojos o la piel queden expuestos tanto a la radiacion directa como a la dispersa. Producto laser clase 4.	Zichtbare en onzichtbare laserstraling! Vermijd blootstelling van oog of huid ann direkte straling of terugkaatsingen daarvan! Klas 4 laser produkt.

CE Declaration of Conformity (Low Emissions)

We,

Spectra-Physics, Inc.
Industrial and Scientific Lasers
1330 Terra Bella Avenue
P.O. Box 7013
Mountain View, CA. 94039-7013
United States of America

declare under sole responsibility that the:

Spitfire Multi-Kilohertz Ti:Sapphire Regenerative Amplifier System with SDG II Controller,

Manufactured after December 31, 1996

meets the intent of "Directive 89/336/EEC for Electromagnetic Compatibility."

Compliance was demonstrated (Class A) to the following specifications as listed in the official *Journal of the European Communities*:

EN 50081-2:1993 Emissions:

EN55011 Class A Radiated
EN55011 Class A Conducted

EN 50082-1:1992 Immunity:

IEC 801-2 Electrostatic Discharge
IEC 801-3 RF Radiated
IEC 801-4 Fast Transients

I, the undersigned, hereby declare that the equipment specified above conforms to the above Directives and Standards.



Bruce Craig
Vice President and General Manager
Spectra-Physics
Laser Group
April 5, 2002
Mountain View, California
USA

CE Declaration of Conformity (Low Voltage)

We,

Spectra-Physics, Inc.
Industrial and Scientific Lasers
1330 Terra Bella Avenue
P.O. Box 7013
Mountain View, CA. 94039-7013
United States of America

declare under sole responsibility that the

Spitfire Multi-Kilohertz Ti:Sapphire Regenerative Amplifier System with SDG II Controller,

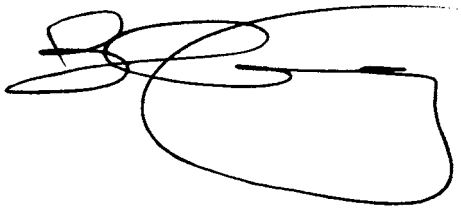
meets the intent of "Directive 73/23/EEC, the Low Voltage directive."

Compliance was demonstrated to the following specifications as listed in the official *Journal of the European Communities*:

EN 61010-1: 1993 Safety Requirements for Electrical Equipment for Measurement, Control and Laboratory use:

EN 60825-1: 1993 Safety for Laser Products.

I, the undersigned, hereby declare that the equipment specified above conforms to the above Directives and Standards.



Bruce Craig
Vice President and General Manager
Spectra-Physics
Laser Group
April 5, 2002
Mountain View, California
USA

Sources for Additional Information

The following are some sources for additional information on laser safety standards, safety equipment, and training.

Laser Safety Standards

Safe Use of Lasers (Z136.1)
American National Standards Institute (ANSI)
11 West 42nd Street
New York, NY 10036
Tel: (212) 642-4900

Occupational Safety and Health Administration (Publication 8.1-7)
U. S. Department of Labor
200 Constitution Avenue N. W., Room N3647
Washington, DC 20210
Tel: (202) 693-1999
Internet: www.osha.gov

A Guide for Control of Laser Hazards
American Conference of Governmental and
Industrial Hygienists (ACGIH)
1330 Kemper Meadow Drive
Cincinnati, OH 45240
Tel: (513) 742-2020

Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (800) 345-2737
Internet: www.laserinstitute.org

Compliance Engineering
70 Codman Hill Road
Boxborough, MA 01719
Tel: (978) 635-8580

International Electrotechnical Commission
Journal of the European Communities
IEC60825-1 Safety of Laser Products—Part 1: Equipment Classification, Requirements and User's Guide
IEC-309—Plug, Outlet and Socket Coupler for Industrial Uses
Tel: +41 22-919-0211
Fax: +41 22-919-0300
Internet: www.iec.ch

Cenelec
European Committee for Electrotechnical Standardization
35, Rue de Stassartstraat
B-1050 Brussels, Belgium
Tel: +32 2 519 68 71
Internet: www.cenelec.org

Document Center, Inc.
111 Industrial Road, Suite 9
Belmont, CA 94002-4044
Tel: (650) 591-7600
Internet: www.document-center.com

Equipment and Training

Laser Safety Guide
Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (407) 380-1553
Tel: (800) 34 LASER
Internet: www.laserinstitute.org

Laser Focus World Buyer's Guide
Laser Focus World
Penwell Publishing
98 Spit Brook Road
Nashua, NH 03062
Tel: (603) 891-0123
Internet: http://lfw.pennet.com/home.cfm

Photonics Spectra Buyer's Guide
Photonics Spectra
Laurin Publications
Berkshire Common
PO Box 4949
Pittsfield, MA 01202-4949
Tel: (413) 499-0514
Internet: www.photonics.com/directory/bg/XQ/ASP/QX/index.htm

Chapter 3 General Description

The *Spitfire* amplifier system contains all the components necessary to amplify low-energy Ti:sapphire laser pulses to energy levels as high as a millijoule. The *Spitfire* amplifier comprises the optical stretcher, the regenerative amplifier and the optical compressor. The femtosecond or picosecond seed pulses to be amplified are provided by a separate mode-locked Ti:sapphire laser system.

The *Spitfire* system also includes the Synchronization and Delay Generator, the *SDG II*, which provides the precise timing required to select pulses for amplification and to eject them from the amplifier. The functions of both the amplifier and *SDG II* are described in this chapter.

Ti:Sapphire

Ti:sapphire is a crystalline material produced by introducing Ti_2O_3 into a melt of Al_2O_3, where Ti^{3+} (titanium) ions replace a small percentage of the Al^{3+} (aluminium) ions. A boule of material is then grown from this melt. The Ti^{3+} ion is responsible for the lasing action in Ti:sapphire. The electronic ground state of the Ti^{3+} ion is split into a pair of vibrationally broadened levels as shown in Figure 3-1.

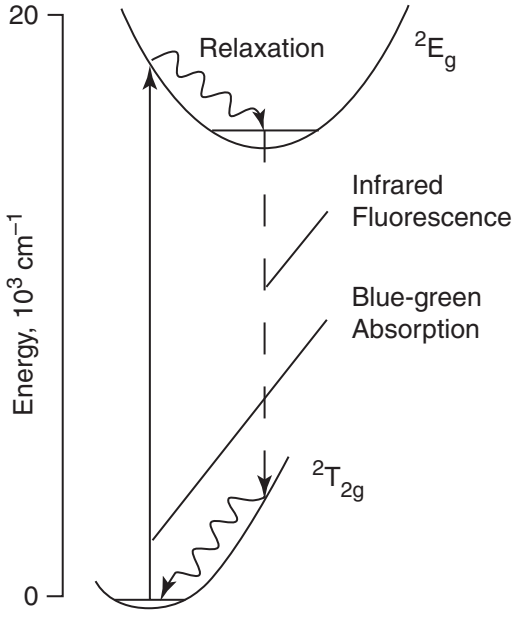


Figure 3-1: Energy Level Structure of Ti:Sapphire

Absorption transitions occur over a broad range of wavelengths from 400 to 600 nm (only one of which is shown in Figure 3-1). Fluorescence transitions occur from the lower vibrational levels of the excited state to the upper vibrational levels of the ground state. The resulting emission and absorption spectra are shown in Figure 3-2.

Although the fluorescence band extends from wavelengths as short as 600 nm to wavelengths greater than 1000 nm, lasing action is only possible at wavelengths longer than 670 nm because the long wavelength side of the absorption band overlaps the short wavelength end of the fluorescence spectrum. Additionally, the tuning range may be reduced by variations in mirror coatings, tuning element losses, pump power and pump mode quality.

Nevertheless, Ti:sapphire possesses the broadest continuous wavelength tuning range of any commercially available laser. As discussed in the following sections, this broad tuning range allows Ti:sapphire lasers to produce and amplify optical pulses of extremely short duration. As a corollary, the same factors that allow Ti:sapphire a broad, tunable wavelength range might also affect the production and amplification of these ultrashort pulses.

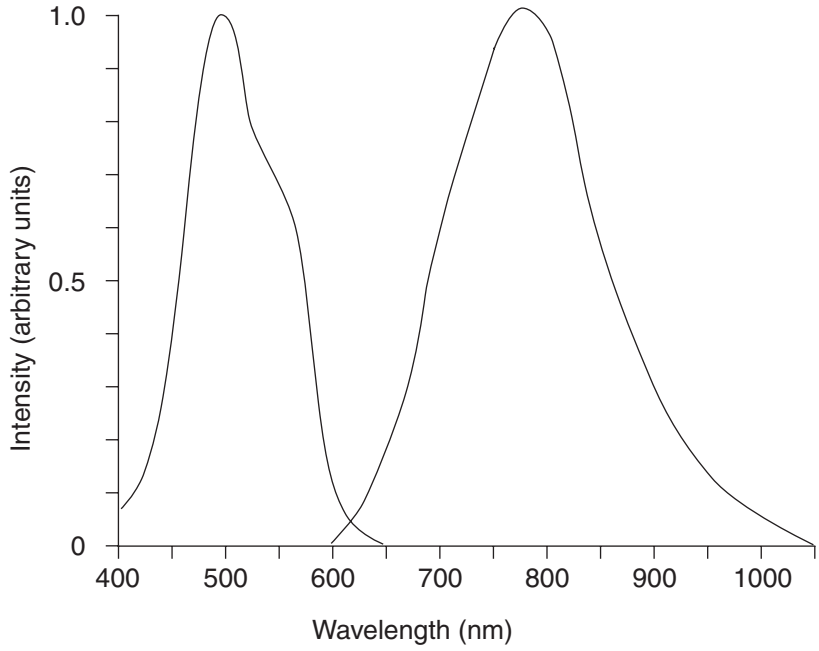


Figure 3-2: Absorption and Emission Spectra of Ti:Sapphire

The Ti:sapphire crystal is highly resistant to thermally induced stress. This resistance allows it to be optically pumped at relatively high average powers without danger of fracture. However, it cannot handle the high peak powers that would result from directly amplifying femtosecond pulses. A technique called Chirped Pulse Amplification, which temporally stretches the pulse prior to amplification and then recompresses it following amplification, circumvents this limitation.

Chirped Pulse Amplification

When an intense beam travels through a Ti:sapphire crystal, it tends to "self-focus." Self-focusing is a nonlinear optical effect in which an intense light beam modifies the refractive index of the material it is passing through, causing the beam to focus and intensify even further. This can potentially result in a run-away condition that causes permanent damage to the crystal. Therefore, self-focusing makes it necessary to limit the peak power of a pulse in the Ti:sapphire crystal to less than 10 GW/cm^2.

Chirped Pulse Amplification (CPA) allows a Ti:sapphire crystal to be used to amplify pulses beyond this peak power, while keeping the power density in the amplifier below the damage threshold of the crystal. CPA is accomplished in three steps. The first step stretches the very short seed pulse that is supplied by a stable, mode-locked picosecond or femtosecond laser. Stretching the pulse (i.e., increasing its duration) reduces its peak power, which greatly reduces the probability of damage to the Ti:sapphire amplifier crystal.

The second step amplifies the stretched pulse: a pump laser provides a synchronous energy pulse to the Ti:sapphire crystal to excite it just prior to the arrival of the stretched seed pulse. The seed pulse causes stimulated emission, which amplifies the pulse at the same wavelength and direction. (This is in contrast to "spontaneous emission" within the gain medium that typically is amplified to become laser output in other systems.)

The third step recompresses the stretched, amplified pulse as close as possible to its original duration.

How It Works

The fundamental relationship that exists between laser pulse width and bandwidth is that a very short pulse exhibits a broad bandwidth, and vice versa. For a Gaussian pulse, this relation is given as

$$d\nu * dt > 0.441 \qquad [1]$$

where $d\nu$ is the bandwidth and dt is the laser pulse width. For example, for a 100 fs duration pulse at $\lambda = 800$ nm, the corresponding bandwidth is more than 9 nm. Therefore, a device that can delay certain frequencies (or wavelengths) relative to others can stretch a short pulse so that it lasts a longer time. Likewise, such a device should also be able to compress a long pulse into a shorter one by reversing the procedure. The phenomenon of delaying or advancing some wavelengths relative to others is called Group Velocity Dispersion (GVD) or, less formally, "chirp."

A pulse is said to have positive GVD, or to be positively chirped, when the shorter (bluer) wavelengths lead the longer (redder) wavelengths. Conversely, if the bluer light is delayed more than the redder light, it has negative GVD or chirp.

For CPA, a combination of dispersive optics are used to form a "pulse stretcher" where low-energy, short-duration pulses can be lengthened by as much as 10^4. Then the energy in these pulses is increased by passing them

through the Ti:sapphire regenerative amplifier. Finally, a set of dispersive optics (similar to those used in the stretcher) are used to form a "pulse compressor" to recompress the pulses to their specified duration. Figure 3-3 illustrates this process.

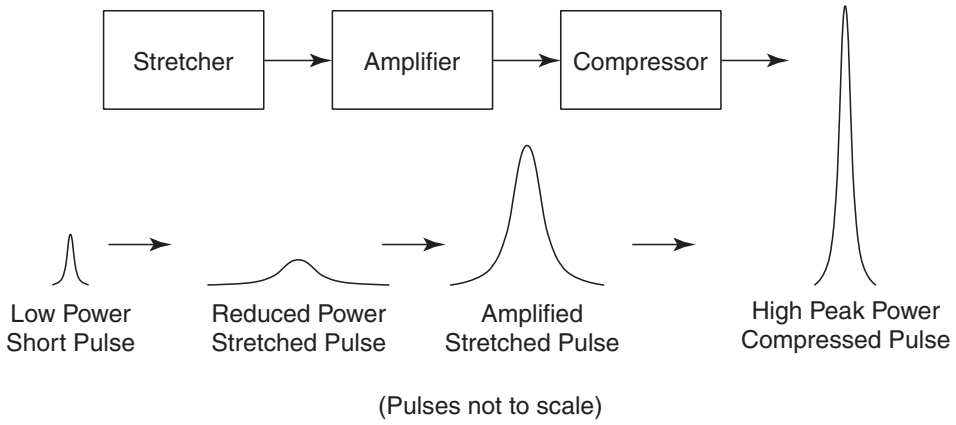


Figure 3-3: The Principle of Chirped Pulse Amplification

Pulse Stretching and Compression

A light pulse incident on a diffraction grating experiences dispersion; that is, its component wavelengths are spatially separated, and so too are its frequency components. The dispersed spectrum can be directed through a combination of optics (usually the same diffraction grating can be used) to send the different frequencies in slightly different directions. Longer (or redder) wavelengths can be made to travel over a longer path than the shorter (or bluer) wavelengths components of the beam, or vice versa. The result is to lengthen the duration of the pulse, which reduces its peak power (it is the same energy under the curve, only spread out more now).

A prism, which is a simpler optic than a diffraction grating, can also be used for these purposes. However because the pulse passes *through* a prism, negative GVD is introduced by the glass or quartz of the prism body — blue frequencies are delayed relative to the red frequencies each time the pulse passes through the prism. Therefore, gratings are the better choice for CPA because they simplify the process of compensating for dispersion caused by other components in the optical path.

The grating and the routing mirrors can be chosen so that, in the stretcher, the bluer frequency components of the spectrum travel further than the redder components, causing the redder frequency components to exit the stretcher first. In the compressor, the spatially spread beam is flipped so that the redder component have to take the long path, thereby allowing the bluer frequencies to catch up. This recompresses the pulse.

Figure 3-4 shows a simplified pulse stretcher. A short pulse is spectrally spread and then, by making one end of the spread pulse travel farther than the other end, the pulse is temporally broadened. The same optical components act as a compressor when the leading component of a temporally stretched pulse is forced to take the longer path, thereby allowing the trail-

ing component to catch up. In the pulse stretcher shown below, the bluer components are forced to take the longer path.

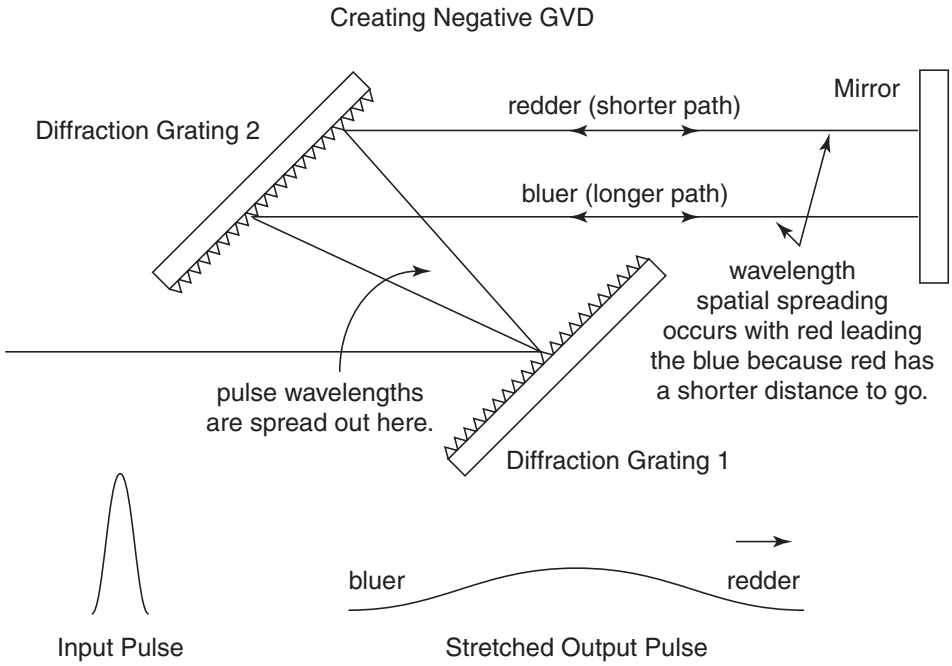


Figure 3-4: Principle of pulse stretching using negative GVD

The Spitfire Pulse Stretcher and Compressor

The *Spitfire* pulse stretcher and compressor make use of some simplifying modifications. Instead of using two gratings for the stretcher, a simple but elegant retroreflector mirror assembly directs the beam back onto a single grating in the stretcher. This avoids the need to match or to precisely align two stretcher gratings. The beam is also multi-passed to achieve greater spectral spread at reduced complexity and cost.

The same design principal is used in the compressor, but in reverse. The result is only two gratings are used in the entire system instead of four, simplifying the alignment and maintenance of the system.

If the input to the *Spitfire* is tuned to a different wavelength, the diffraction grating in the stretcher will cause the beam to move, and the grating must be rotated to realign the stretcher. Naturally, the compressor grating must be rotated by exactly the same amount to ensure optimum pulse compression.

To make this adjustment simple, the *Spitfire* stretcher and compressor gratings are arranged back-to-back on the same mount so that only one adjustment is necessary to accommodate a change in wavelength.

Note

The gratings for the *Spitfire 50FS* model are mounted separately and, therefore, must be adjusted individually. Refer to Chapter 7 for details.

The stretcher and compressor occupy a single chamber and are separated from the amplifier by an air baffle that minimizes air currents through the stretcher and compressor. The compressor uses a horizontal retroreflector to flip the red and blue components so that the bluer wavelengths are now forced to take the longer path. This allows the redder wavelengths to catch up and reduce the pulse duration to close to its original length.

The retroreflector in the compressor is mounted on a track for easy translation in the direction along the beam path. This fine adjustment is used to compensate for small, routine changes in dispersion that take place in the amplifier cavity. Translation control is provided by a motion controller and dc motor.

The design details of the gratings and their optical configuration in a CPA system depend upon, among other factors, the duration of the seed pulses and output pulses. Longer duration pulses have a correspondingly narrower spectrum of wavelengths, and so require a higher density of rulings for the diffraction gratings to achieve an adequate degree of dispersion and stretching.

Each *Spitfire* model has its own stretcher/compressor design. The *Spitfire F, P, PM* and *USF* each use gratings that are designed for their specific pulse lengths, and from the nature of the grating diffraction, this means they each have their back-to-back gratings set at different angles to the beam path. There are some other individual differences but, overall, these stretcher/compressor designs are similar.

The *Spitfire 50FS* differs from the other models in the layout of its stretcher and compressor. The *Spitfire PM* includes a masking element to change the bandwidth of the seed pulses. The designs of the stretcher/compressor combinations for both of these models is discussed in further detail below.

The Spitfire 50FS Compressor/Stretcher Design

Because the GVD phenomenon described above is not a simple linear effect, extra consideration must be given to a design intended to produce the shortest possible pulses. The frequency components of a pulse traversing an optical system experience dispersion that depends upon the square of the frequency. In addition, dispersion also results from higher order powers of the frequency. For pulses around 100 fs or longer, this higher-order dispersion is small enough so that it is adequately compensated by the robust back-to-back design of the *Spitfire* stretcher/compressor.

However, for pulses of extremely short duration, the higher-order dispersion becomes large enough that additional compensation is required. The *Spitfire 50FS* compensates for this higher-order dispersion by using a “mixed” stretcher and compressor design—the groove densities of the gratings are different. This requires that the gratings be adjusted independent of one another. To provide this flexibility, the *Spitfire 50FS* gratings are installed on separate mounts.

The Spitfire PM Compressor/Stretcher Design

Some applications require amplified picosecond pulses rather than femtosecond pulses. The *Spitfire P* system produces picosecond pulses from picosecond duration seed pulses, such as those produced by the picosecond version of the Spectra-Physics *Tsunami*. The operation of the *Spitfire P* is based on the principals described in the sections that began this chapter.

In many installations, however, only a mode-locked femtosecond laser (such as the femtosecond *Tsunami*) is available to seed the *Spitfire* amplifier. The *Spitfire PM* ("pico-mask") system allows you to produce amplified picosecond pulses using the femtosecond seed laser. This configuration has specific advantages for certain applications, such as pumping an optical parametric amplifier. The *Spitfire PM* converts the pulse spectrum of the femtosecond seed pulses into a spectrum that is equivalent to that produced by a picosecond seed laser.

Recall the relationship between laser pulse width and bandwidth described in "Chirped Pulse Amplification": a very short pulse exhibits a broad spectrum; longer pulses exhibit narrower spectra. Since the pulse stretcher works by spatially separating the spectrum of the seed pulses, the bandwidth of these pulses will be reduced if part of this spectrum is discarded. When the pulse is later compressed, its duration will be longer than if the entire spectrum had been preserved.

The *Spitfire PM* accomplishes this in a conceptually straightforward manner. A femtosecond seed pulse enters a stretcher that uses a grating designed for picosecond operation. The spatially spread pulse is then directed onto an aperture that is precisely aligned to mask part of the spectrum that reflects from the grating, and a portion of the spectrum is allowed to pass through. The bandwidth of this stretched pulse is thereby reduced to that produced by a picosecond seed pulse.

If these reduced-bandwidth pulses were allowed to pass through a stretcher designed for femtosecond pulses, optical damage might result. The bandwidth of these masked pulses is much narrower than the bandwidth of femtosecond pulses—about ½ nanometer as opposed to 10 nanometers. (This is also true for the picosecond seed pulses in the *Spitfire P* amplifier.)

Picosecond pulses require a greater degree of dispersion to produce spatial and, hence, temporal, separation. This is achieved by increasing the path length of the beam in the stretcher and increasing the ruling density of the stretcher grating. After amplification, the stretched pulse is directed into the compressor, which is configured just as it would be for a picosecond seed laser (especially in the choice of compressor grating ruling), and amplified picosecond pulses are the result.

A detailed description of the optical layout of the *Spitfire PM* is given in Chapter 7. Procedures for converting a *Spitfire F* or *USF* system to *Spitfire PM* operation (or vice-versa) are given in Appendix B, "Changing to/from PicoMask Operation."

Pulse Selection and Pockels Cells

Once the pulses leave the stretcher, selecting a pulse for retention in the amplifier cavity is accomplished by exploiting its polarization characteristics and by using Pockels cells to control this polarization. A Pockels cell is an electro-optic device that, without an applied voltage, has essentially no effect on light transmitted through it. With an applied voltage, however, the crystalline material in a Pockels cell acts as a ¼ waveplate that rotates the polarization of transmitted light by 45° each time a pulse passes through it.

If a light beam passes through an active Pockels cell twice (passes through the cell and is then reflected back through it again), the polarization of the beam is rotated by 90°, or from horizontal to vertical, or vice versa. However, in order for this to work well, the Pockels cell must be properly aligned with no voltage applied. Likewise the applied voltage must be calibrated to achieve the precise degree of polarization rotation.

The input Pockels cell is paired with a passive ¼ waveplate, and the optical path is designed so that the beam makes a double pass through this combination. When the cell is off, the double pass through the passive ¼ waveplate will flip the beam polarization 90°; when the cell is on, the beam experiences a double pass through two ¼ waveplates, leaving its polarization unchanged.

The *Spitfire* cavity is designed so that horizontally polarized light remains trapped in the cavity and is amplified. Details of how the input Pockels cell combines with the cavity optics to select a pulse for amplification are given in Chapter 7.

The output Pockels cell works in conjunction with the polarizer in the amplifier cavity to release an amplified pulse at a time determined by the *SDG II*. Details of how the timing is set for ejecting an amplified pulse are given in the section “The Synchronization and Delay Generator (SDG II)” on page 3-10.

One measure of the quality of pulse selection is given by the contrast ratio—the factor by which the amplifier output power exceeds the power in spurious pulses which are always present to some degree before or after the main, “true” pulse. The value of the contrast ratio is determined by the quality of the ¼ waveplates, the activation time of the Pockels cells (their intrinsic birefringence) and their drive electronics.

The components selected for the *Spitfire* that affect contrast ratio are of the highest quality available. As a consequence, the limiting factor for contrast ratio is the natural birefringence of Pockels cells. This birefringence results in an optical rise time that is less than a nanosecond. The net effect is high contrast ratios and excellent suppression of spurious pulses.

Note



The 3500 V applied to the *Spitfire* Pockels cells is provided by two high-voltage power supplies. For 1 kHz systems, these power supplies are in the *SDG II*. An auxiliary high-voltage power supply is provided for 5 kHz systems. The Pockels cells themselves are in the *Spitfire* amplifier.

Regenerative Amplification

A typical laser amplifies the spontaneous emission randomly present in its own gain medium in order to initiate lasing. Regenerative amplifiers, on the other hand, are designed to recirculate and amplify low-energy laser pulses from a separate "seed" laser and are an efficient means of generating high peak-power pulses. Thus, instead of allowing the energy in the amplifier crystal to escape as random spontaneous emission, these seed pulses (having an energy that exceeds the spontaneous emission energy) are selectively amplified. The *Spitfire* can be thought of as a Q-switched, cavity-dumped Ti:sapphire laser that is configured to operate as an amplifier. Here is how it works:

As explained earlier, Ti:sapphire has a broad gain bandwidth that is necessary for the production and amplification of sub-picosecond pulses. In addition, a Ti:sapphire amplifier has a high saturation threshold that makes it possible to extract relatively high energies from a system of moderate size.

A single pass of a very low-energy sub-picosecond pulse through a Ti:sapphire amplifier will increase the pulse energy typically by a factor of about 3 or 4. However, the stimulated emission that provides this gain draws down the population inversion in the gain media only a small amount in a single pass, thus allowing the gain media to remain well below the threshold at which stimulated emission will reverse the population inversion (that is, saturate the gain) and amplification stops. In short, after a single pass of a low-energy pulse, there is still a lot of gain left in the amplifier for more passes.

The *Spitfire* cavity is designed to first select and then optically confine an individual pulse from the train of mode-locked seed pulses that have already been lengthened in duration in the stretcher. Reducing the repetition rate from the megahertz mode-locked pulse train to kilohertz rates enables the gain of the amplifier to be concentrated in fewer pulses, thus producing more energy per pulse.

Immediately prior to passing the selected pulse through the Ti:sapphire crystal for amplification, the crystal is exited to population inversion by a high-energy pulse from a separate pump laser. The selected pulse is then passed through the crystal 20 or more times until the stimulated emission (the pulse energy level) is high enough to completely eliminate the population inversion. Having thus saturated the gain, i.e., absorbed all the energy available, the pulse is ejected into the compressor.

Typically, an input pulse of only a few nanojoules of energy may be amplified to roughly a millijoule using a single Ti:sapphire crystal, and multiple passes through the regenerative amplifier can result in an energy amplification greater than 10^6 at the output of the compressor. When the compressor restores the short duration of the pulse, the amplified energy results in correspondingly amplified peak power.

Note 

As part of the alignment procedure, the *Spitfire* is sometimes operated as a laser rather than as a regenerative amplifier.

The Synchronization and Delay Generator (SDG II)

The Synchronization and Delay Generator, or *SDG II*, provides the timing needed to synchronize the Pockels cells to the passage of the pulses through the amplifier. This allows the Pockels cells to first capture pulses and then, later, to direct them into the compressor. This timing includes synchronization to the seed and pump lasers. The *SDG II* also provides an adjustable delay based on the output of the *Spitfire* that allows laboratory instruments to be synchronized to the arrival of pulses at the target.

Immediately after the Ti:sapphire rod is excited by a pulse from the pump laser, the input Pockels cell confines a selected pulse in the amplifier and sends it into the rod for amplification. The input Pockels cell therefore must be synchronized to the mode-locked pulse train after the next available pump pulse, and remain synchronized after each pump pulse.

To achieve this, the input Pockels cell is locked to the RF signal generated by the modelocker in the seed laser. Additionally, the Pockels cell firing phase (the delay) is adjustable to allow the synchronization to be optimized. This ensures that the input Pockels cell fires only after the selected pulse has passed completely through it.

The output Pockels cell ejects the amplified pulse into the compressor. Following the synchronization of the input Pockels cell, there is a delay before the output cell is activated to ensure the captured pulse is released at optimum amplification. This delay is adjustable from 0 to 1275 ns, which allows the pulse to complete the amplifier cavity path an integral number of times.

The *SDG II* is first triggered by a TTL positive edge pulse provided by the pump laser. It then produces separate triggers with adjustable delays for both Pockels cells. OUT 1 DELAY on the front panel connects to the input Pockels cell; OUT 2 DELAY connects to the output Pockels cell.

Delay adjustment is via the corresponding knobs on the front panel, and each delay is displayed in nanoseconds above each knob. Adjusting OUT 3 DELAY allows the user to synchronize target or monitoring devices to the *Spitfire* output pulse. As a simplified example, OUT 3 DELAY can be used to provide horizontal triggering for an oscilloscope.

Note



The repetition rate of Pockels cell switching (and hence the repetition frequency of the *Spitfire* output) is dependent on the repetition rate of the input trigger from the pump laser.

For 1 kHz systems, the *SDG II* also contains the high-voltage power supplies used to power the Pockels cells (5 kHz systems use a separate high-voltage power supply). The *SDG II* contains the control and the signals for the Bandwidth Detector (BWD).

The control of the BWD is explained in Chapter 4, “Controls, Indicators and Connections,” along with instructions for operating the *SDG II*.

Specifications

The *Spitfire* amplifier systems are available in a number of configurations. The tables below show the configurations covered in this manual.

Table 3-1: *Spitfire* Specifications[1] by Model

Amplifier Model[2]	Output Energy[3] using these pump lasers		Pulse Width[4]	Pre-Pulse Contrast Ratio[5]	Wavelength[6] (nm)
	Evolution	Evolution X			
F-1K	750 µJ	1 mJ	<130 fs	1000:1	750–900
F-5K	200 µJ	300 µJ	<130 fs	500:1	750–900
P-1K	750 µJ	1 mJ	<2 ps	1000:1	750–900
P-5K	200 µJ	300 µJ	<2 ps	500:1	750–900
PM-1K	750 µJ	1 mJ	1–2 ps	1000:1	750–900
PM-5K	200 µJ	300 µJ	1–2 ps	500:1	750–900
USF-1K	500 µJ	750 µJ	<90 fs	1000:1	750–900
USF-5K	150 µJ	225 µJ	<90 fs	500:1	750–900
50 FS-1K	500 µJ	700 µJ	<50 fs	1000:1	780–820
50 FS-5K	150 µJ	200 µJ	<50 fs	500:1	780–820

[1] *Due to our continuous product improvement program, specifications may change without notice. Specifications listed on the purchase order supersede all other published specifications.*

[2] *Designators "1K" and "5K" refer to repetition rates of 1 kHz and 5 kHz, respectively. If optimum performance is required at more than one repetition rate, an additional optic set is required. Any system can be operated with the same energy per pulse at reduced repetition rates through the divide-down electronics on the SDG II.*

[3] *Output energy per pulse: applies between 780 and 800 nm. For higher energy output systems, please contact Spectra-Physics.*

[4] *Pulse width applies at the peak wavelength and requires the seed laser performance specified for the Amplifier model. A Gaussian pulse shape (0.7 deconvolution factor) is used to determine the pulse width (FWHM) from an autocorrelation signal as measured with a Spectra-Physics Model SSA (current version).*

[5] *Contrast ratio is defined as the ratio between the peak intensity of the output pulse to the peak intensity of any pulse that occurs more than 1 ns before the output pulse. The contrast ratio for any pulse more than 1 ns after the output pulse ("post-pulse contrast ratio") is > 100:1. For higher performance, please contact Spectra-Physics.*

[6] *Wavelength: the system is tunable from 750 to 840 nm without an optics change. To cover the 840–900 nm region, separate optics are required for both 1 K and 5 K systems. For other wavelengths and for second and third harmonic generation, please contact Spectra-Physics.*

Table 3-2: *Spitfire* Specifications Common to All Models

Beam Diameter[1]	Beam Divergence[2]	Transform Limit[3]	Energy Stability[4]	Output Polarization
7 mm	<1.5	<1.5	<±3%	horizontal

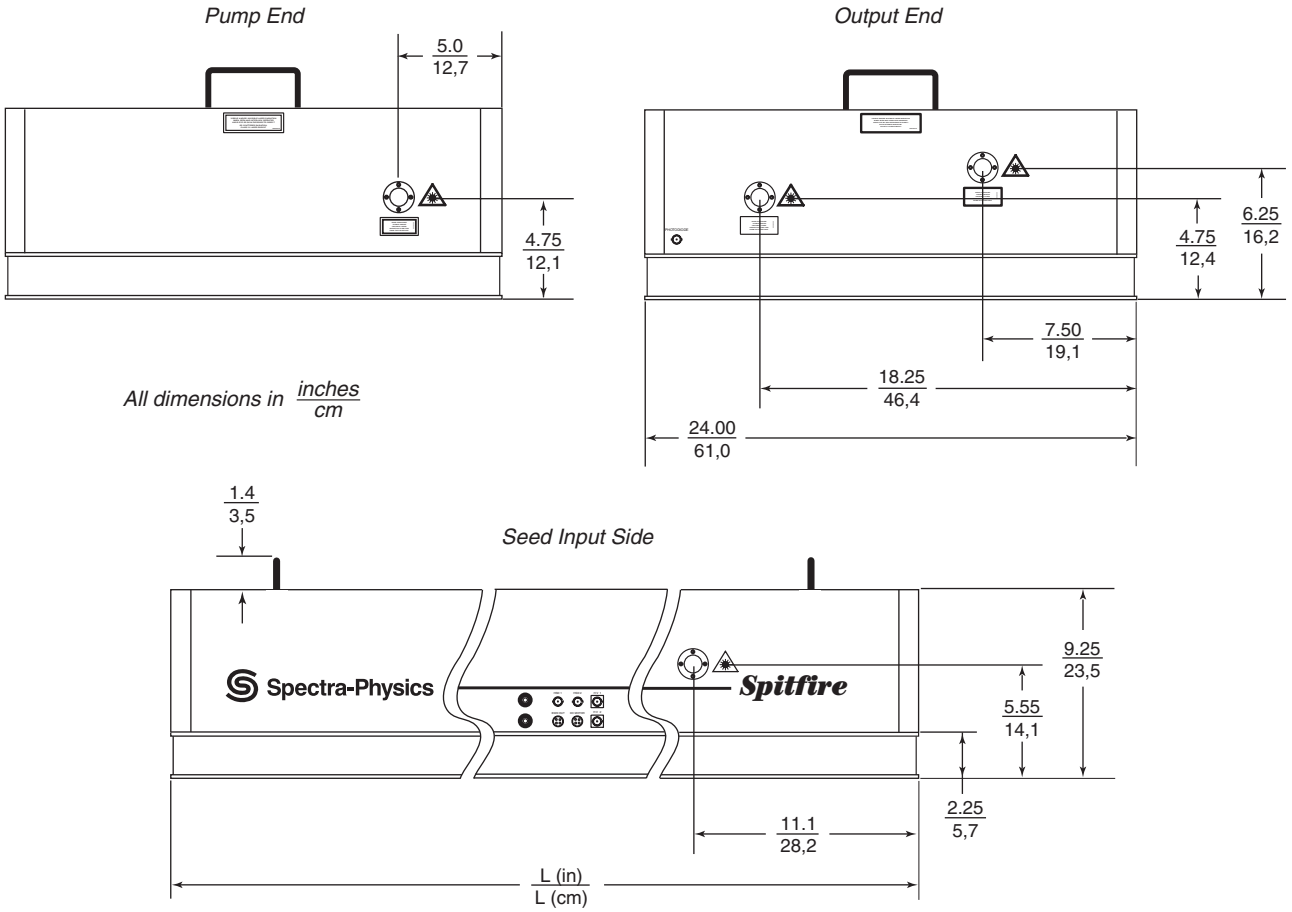
[1] *Nominal beam diameter at $1/e^2$ points.*

[2] *Beam divergence as a multiple of diffraction limit.*

[3] *Assuming seed pulses are transform-limited Gaussian temporal pulses.*

[4] *Applies at peak wavelength between 780 and 800 nm.*

Outline Drawings



Model	L (in)	L (cm)
Spitfire F, P, PM & USF	48.0	121,9
Spitfire 50FS	60.0	152,4

Figure 3-5: *Spitfire* Outline Drawing

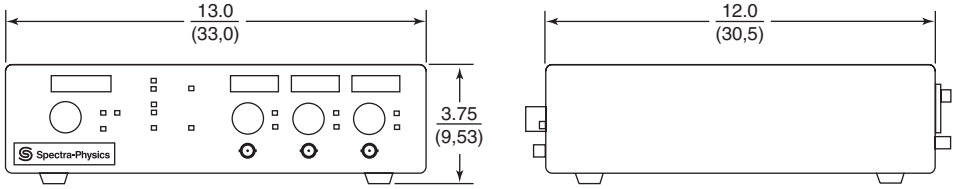


Figure 3-6: *SDG II* Outline Drawing

Chapter 4 Controls, Indicators and Connections

This chapter describes the controls, indicators and connections needed to operate the *Spitfire* system. It describes the external panels of the *Spitfire* amplifier, the synchronous delay generator (*SDG II*) and auxiliary connections.

Occasionally, troubleshooting or optimizing system performance may require adjustment of the optical components inside the *Spitfire* amplifier. The internal adjustments for aligning the optical path inside the *Spitfire* are described separately in Chapter 7.

The *Spitfire* can be controlled by a computer via the RS-232 interface on the *SDG II*. Appendix A provides information regarding the command language used by this system.

Spitfire Head External Controls

Pump Input End Panel

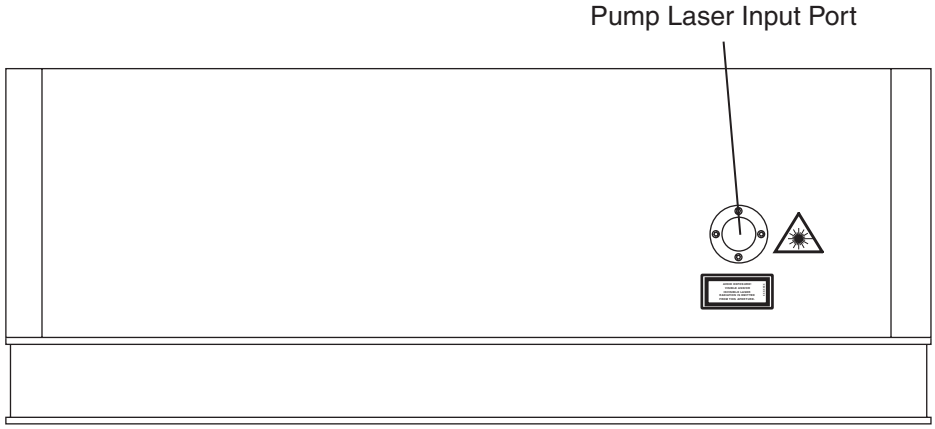


Figure 4-1: *Spitfire* Panel, Pump Input End

Pump laser input port—is the input port for the beam from the pump laser (e.g., a Spectra-Physics *Evolution* Q-switched laser).

Seed Input Side Panel

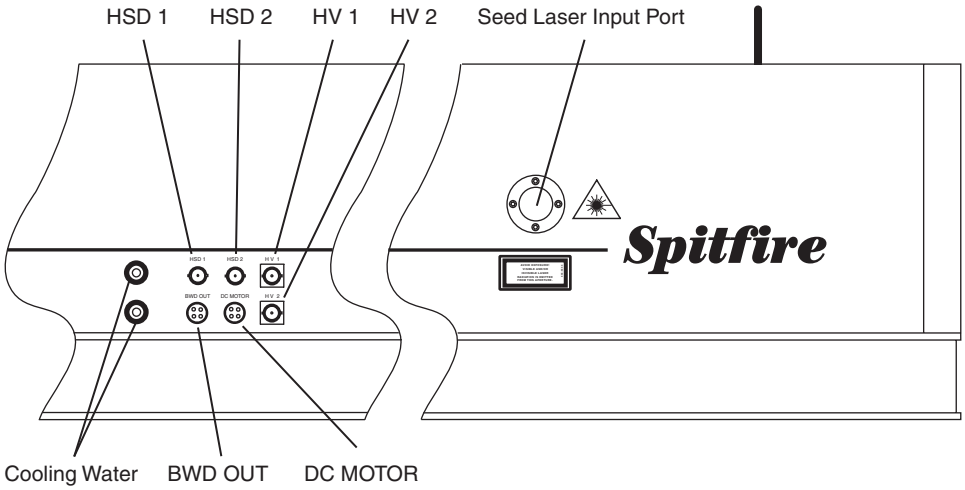


Figure 4-2: *Spitfire* Panel, Seed Laser Input Side

Seed laser input port—provides an input port for the seed beam (*Tsunami* or *Mai Tai* mode-locked laser).

HV 1 connector (MHV)—**HIGH VOLTAGE**—connects via a high-voltage cable to the 1–6 kVdc H.V. 1 output connector on the back of the *SDG II* (1 kHz systems) or to the auxiliary power supply (5 kHz systems) for driving the input Pockels cell.

HV 2 connector (MHV)—**HIGH VOLTAGE**—connects via a high-voltage cable to the 1–6 kVdc H.V. 2 output connector on the back of the *SDG II* (1 kHz systems) or to the auxiliary power supply (5 kHz systems) for driving the output Pockels cell.

HSD 1 connector (BNC)—connects to the OUT 1 DELAY connector on the front of the *SDG II* for triggering the input Pockels cell.

HSD 2 connector (BNC)—connects to the OUT 2 DELAY connector on the front of the *SDG II* for triggering the output Pockels cell.

BWD OUT—connects to the 4-pin BWD connector on the back of the *SDG II*.

DC MOTOR input connector—connects to the motor controller (provided with the system) that drives the micrometer motor, which sets the length of the compressor. Refer to "Motion Controller" below.

Cooling water connections—provide cooling water for the amplifier rod. Water is shared serially downstream from the seed laser (*Mai Tai* or *Tsunami*). Either connector may be used as the IN or OUT connection for the water flow.

Output End Panel

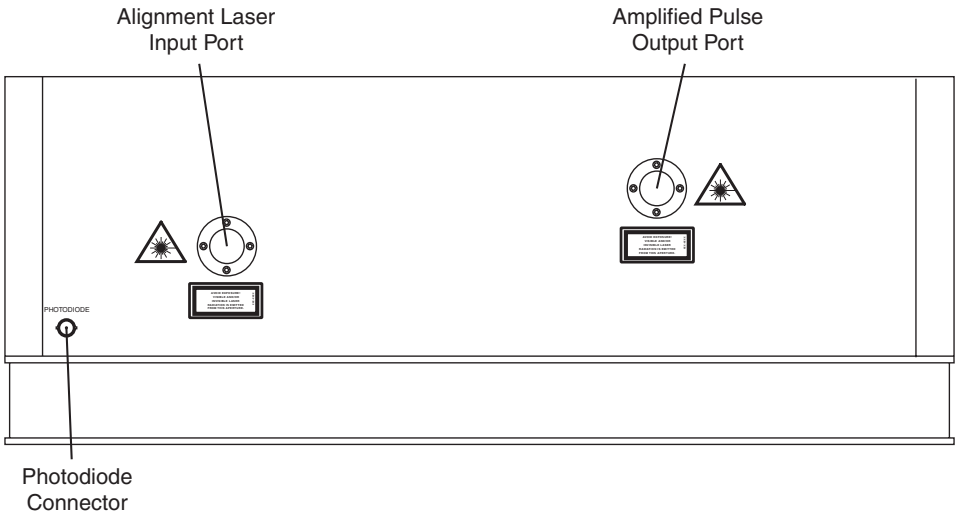


Figure 4-3: *Spitfire* Panel, Output End

Photodiode connector—provides connection for the high-speed photodiode that samples the intracavity signal of the *Spitfire*. The signal can be monitored using a high-speed oscilloscope or spectrometer.

Amplified pulse output port—is the exit port for the amplified pulse.

Alignment laser input port—allows the beam of an alignment (HeNe) laser to be injected into the amplifier optical train without removing the output end panel. To use this port, remove the photodiode module inside the amplifier.

The photodiode detector module resides directly behind the end mirror of the amplifier.



The alignment laser input port must be closed while either the *Spitfire* or the pump or the seed lasers are operating.

The Synchronous Delay Generator

The Synchronous Delay Generator (*SDG II*) controls the selection of pulses from the seed laser and the repetition rate of the pulsed output of the *Spitfire*. It acts as a counter that counts and then selects mode-locked seed pump pulses at either the 1 kHz or the 5 kHz amplifier rate.

The *SDG II* also synchronizes the seed pulses with pulses from the pump laser—it captures the next seed pulse while the laser rod is still excited by the pump pulse. It does this by providing an adjustable delay (in nanoseconds) that the amplifier input Pockels cell can be set to in order to capture the pulse.

The second adjustable delay controls the output Pockels cell to eject the pulse into the compressor after it has been amplified. The *SDG II* allows the output repetition rate to be reduced from its pre-set value by dividing

the input synchronization signal from the pump laser. Preset integer divider values are provided.

The third adjustable delay provides a trigger for laboratory equipment such as the horizontal sweep of a high-speed oscilloscope.

The *SDG II* also contains the high-voltage power supplies for driving the Pockels cells for 1 kHz systems. 5 kHz systems use an additional, separate high voltage supply. The drivers themselves are located in the *Spitfire* below the regenerative amplifier cavity.

RS-232 control of the *SDG II* is described in Appendix A.

Front Panel

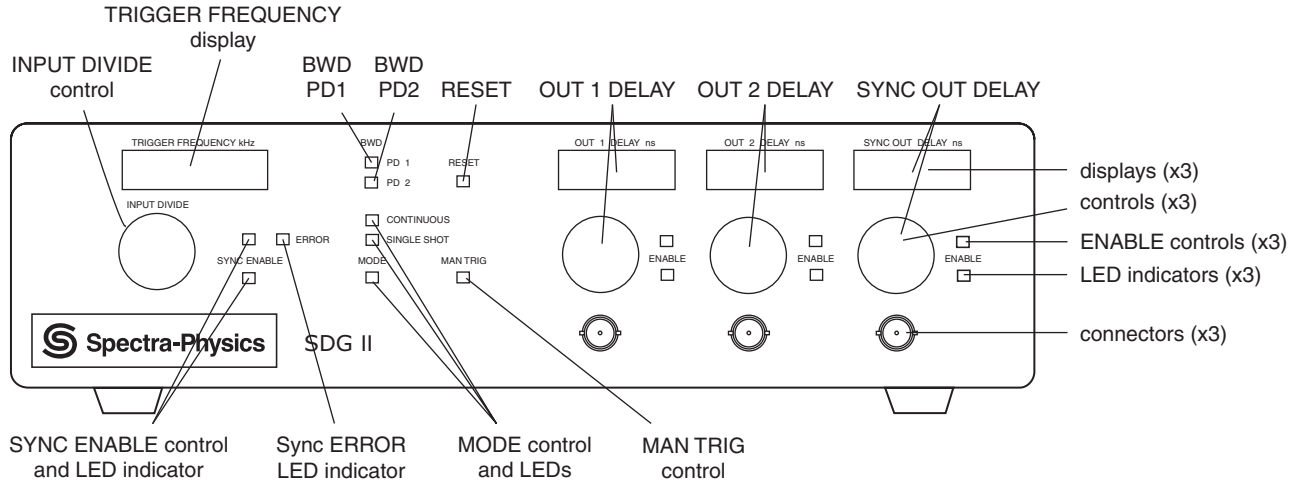


Figure 4-4: *SDG II* Front Panel

TRIGGER FREQUENCY display—shows the output frequency (in kHz) set for the *Spitfire*.

INPUT DIVIDE control—allows the output frequency of the *SDG II* to be reduced by integer divisors (e.g., ÷ 2, ÷ 3, etc.). This allows the output pulse rate of the *Spitfire* to be changed without changing the repetition rate of either the pump laser or the seed laser, which might affect the stability of those lasers.

The largest division factor available corresponds to the reduction of the output to a 1 Hz repetition rate. Thus the largest factor for a 1 kHz system is 1000; the largest factor for a 5 Khz system is 5000. The reduction factor is not shown; only the actual output repetition rate is displayed.

SYNC ENABLE control—selects synchronized (LED is on) or unsynchronized (LED is off) mode. If both the LED and error lamp are on, the sync source is absent or the seed laser has stopped modelocking. Pressing the SYNC ENABLE button again (turning off the LED) will correct the error condition, but it will also disable the synchronization function of the *SDG II*.

Synchronized mode allows the sync outputs to fire based on the current pump laser delay setting (OUT 1 DELAY) and the next available seed pulse.

It provides a way to fire the input Pockels cell based on sync signals from two circuits: the pump laser Q-switch signal and the seed laser pulse train.

SYNC ERROR indicator—when on and the *SDG II* is in synchronized mode, indicates the sync signal is absent or the seed laser is not mode-locked.

BWD (PD_1, PD_2 and RESET)—see "Bandwidth Detector" on page 4-6.

MODE control—selects CONTINUOUS repetition rate firing (based on input trigger) or SINGLE SHOT firing (the corresponding LED turns on)

MAN TRIG control—causes the three output triggers to fire a single pulse when the firing mode is set to SINGLE SHOT and the button is pressed.

ENABLE controls (3)—turn the three adjustable output trigger signals on and off. If a signal is enabled, its corresponding LED is illuminated. When disabled, only that output is deactivated; the other outputs remain active.

OUT 1 DELAY display, control and connector—the display shows the selected delay (0 to 1275 ns) between the pump laser Q-switch sync signal and the time the input Pockels cell is turned on to capture the current seed pulse in the *Spitfire* amplifier.

The control knob adjusts the delay in 250 ps increments, or 10 ns increments if the knob is pushed in during adjustment. The corresponding BNC connector connects to the *Spitfire's* HSD 1 TRIG BNC connector. This provides a low-voltage sync signal to the high-voltage driver, which turns on the input Pockels cell to capture the current seed pulse.

OUT 2 DELAY display, control and connector—the display shows the selected delay (0 to 1275 ns) between the pump laser Q-switch sync signal and the time the output Pockels cell is turned on to eject the amplified pulse into the compressor. This delay *must* be *greater than* the setting for OUT 1 DELAY.

The control knob adjusts the delay in 250 ps increments, or 10 ns increments if the knob is pushed in during adjustment. The corresponding BNC connector connects to the *Spitfire's* HSD 2 TRIG BNC connector. This provides a low-voltage sync signal to the high-voltage driver, which turns on the output Pockels cell to eject the amplified pulse.

SYNC OUT DELAY display, control and connector—the display shows the selected delay (0 to 1275 ns) between the time the output Pockels cell is fired and the time the user can send a trigger signal to a device (such as an oscilloscope) that is part of the target apparatus.

The control knob adjusts the delay in 250 ps increments, or 10 ns increments if the knob is pushed in during adjustment. The corresponding BNC connector connects to the user's oscilloscope for monitoring pulses, or to other apparatus of the target or data acquisition system.

Bandwidth Detector

The Bandwidth Detector (BWD) protects the regenerative amplifier optics from damage if the stretcher cannot adequately reduce the peak power of the seed pulses before they are amplified. This can happen, for example, if a portion of the beam in the stretcher is blocked.

When the seed laser is stable and properly mode-locked, the BWD permits the *SDG II* to function normally. When the BWD senses a lack of signal, a relay will disable the trigger signal that fires the Pockels cells. No pulses are selected for amplification, thus protecting the optical components.

The BWD relies on the signals from two fast photodetectors placed behind the tall stretcher end mirror. This mirror transmits about 5% of the incident light to the detectors. If the signal from either detector falls below a threshold (factory set for each version of the *Spitfire)*, the BWD is activated.

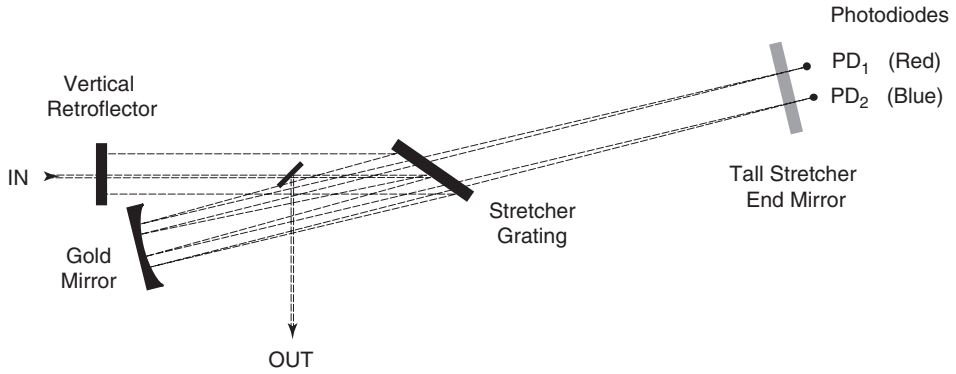


Figure 4-5: Optical Design of the BWD (compressor components are not shown for clarity)

The following indicators and connectors for the BWD are on the *SDG II*:

PD_1, PD_2 indicators ***(front panel)***—when both lamps are on, indicate the stretcher is spreading the seed pulse spectrum properly on the tall stretcher end mirror. PD_1 represents the red end of the spectrum; PD_2 represents the blue end. If a lamp is off, the corresponding photodetector is receiving a signal below threshold.

RESET button ***(front panel)***—when pressed, resets the relay and resumes *Spitfire* amplification after the underlying problem is resolved and both BWD lamps are on.

BWD connector (4-pin, 12 mm) ***(back panel)***—connects to the BWD photodiodes via a similar connector on the *Spitfire.*

BWD ON switch ***(back panel)***—when in the *down* position, disables the BWD and allows the amplifier to function regardless of spectrum spread.

Warning! 

Disabling the BWD can result in permanent damage to the *Spitfire*.

Back Panel

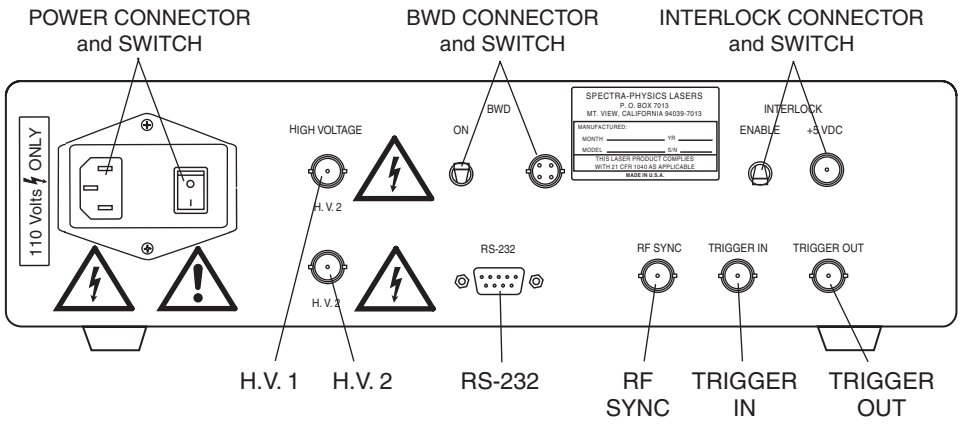


Figure 4-6: *SDG II* Back Panel

Power connector and switch (110/220 Vac)—are the primary power input for the *SDG II*. The unit includes EMI protection, a ½-amp fuse and an on/off switch.

HIGH VOLTAGE (HV1, HV2) connectors—provide 1–6 kVdc output for 1 kHz systems via high-voltage cables to the HSD1 and HSD2 connections on the *Spitfire* for the two Pockels cells.

Note

Voltage for 5 kHz systems is supplied by an auxiliary power supply, and the HV1 and HV2 connectors on the *SDG II* are not used on these systems. Cap these connectors if a 5 kHz system is used.

BWD (ON switch and 4-pin connector)—see "Bandwidth Detector" on page 4-6.

RS-232 connector—provides attachment to a serial connection on a computer for controlling the *SDG II* remotely. Refer to Appendix A for information on the computer control language used with this system.

RF SYNC connector—connects via a high-speed cable to the modelock synchronization output on the seed laser. If a *Mai Tai* or *Tsunami* is used, connect to the 40 MHz output connector (refer to the appropriate user's manual). Jitter is specified at <250 ps; input impedance is 1 MΩ, internally switchable.

TRIGGER IN connector—accepts TTL-compatible, 0–50 kHz input from the Q-switch synchronization output of the pump laser. If a Spectra-Physics *Evolution* pump laser is used, connect to the SYNC OUT connector on the front panel of the power supply. Input impedance is 50 Ω, internally switchable.

TRIGGER OUT connector—provides a 200 ns fixed output trigger signal. The input pulse trigger to the *SDG II* produces this TRIGGER OUT signal and applies it to the three adjustable outputs on the front panel.

INTERLOCK ENABLE switch — enables or disables the +5VDC connector. When the switch is *up,* the connector is functional and the center pin of the BNC is grounded. When the switch is *down,* the connector is disabled.

+5VDC connector (input) — accepts an input signal from a safety interlock switch provided by the user; for example, a switch that senses when a simple closed circuit has opened. If this connector is enabled and the safety interlock switch opens, OUT 1 DELAY and OUT 2 DELAY will be disabled.

Warning!

The use of the +5VDC connector as a safety switch *will not* disable the pump or seed lasers. These lasers have their own safety interlocks. Please refer to their user's manuals. If purchased from Spectra-Physics, these manuals are included with your system.

Motion Controller

The Motion Controller provides translation control of the horizontal retroreflector assembly in the compressor. Moving this mount changes the length of the beam path in the compressor and provides the fine adjustment needed to compensate for small changes in the dispersion that take place in the amplifier cavity.

The Motion Controller connects to the 12 mm, 2-pin connector on the *Spitfire*.

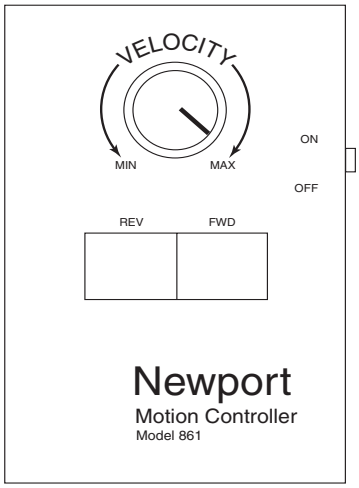


Figure 4-7: Motion Controller (model may vary)

VELOCITY control — sets the speed for the compressor motor micrometer when either the REV or FWD buttons are pushed.

REV button — moves the stretcher to shorten the beam path in the compressor.

FWD button — moves the stretcher to lengthen the beam path in the compressor.

ON/OFF switch — turns the controller on and off. To save the battery, always leave the switch in the OFF position when the controller is not in use.

Chapter 5 Preparing for Installation

Caution! 

Call your Spectra-Physics service representative to arrange an installation appointment, which is part of your purchase agreement. *Allow only authorized Spectra-Physics representatives to install your Spitfire system.* You will be charged for repair of any damage incurred if you attempt to install the *Spitfire* yourself, and such action may void your warranty.

System Components

Because a typical *Spitfire* installation requires both a pump laser and a seed laser in addition to the *Spitfire*, some planning is required before beginning installation. Typical system components include:

- *Evolution* a multi-kilohertz, intracavity doubled, diode-pumped Nd:YLF pump laser

and a

- *Mai Tai* femtosecond Ti:sapphire, modelocked seed laser (this system includes its own internal diode-pumped, CW pump laser)

or a

- *Tsunami* femtosecond or picosecond Ti:sapphire, mode-locked seed laser

and a

- *Millennia* diode-pumped, CW laser for pumping the *Tsunami.*

Note 

Although not recommended, it is possible to use other seed or pump lasers as components in a *Spitfire* system. In particular, seed lasers other than the *Mai Tai* or *Tsunami* will likely require a pre-collimation to avoid the introduction of spatial chirp in the stretcher.

Pump Laser

The *Spitfire* is designed for optimum performance when pumped by a Spectra-Physics *Evolution* laser, a frequency-doubled Nd:YLF laser. Care has been taken to match the *Spitfire* optics to this pump laser, especially with regard to wavelength, beam diameter, divergence and the ability to focus the input beam into the *Spitfire* Ti:sapphire rod.

We recommended the *Spitfire* be pumped only with a Spectra-Physics laser. If this is not the case, your *Spitfire* warranty may be voided unless prior written approval is obtained from Spectra-Physics.

The pump laser must meet the following specifications:

Table 5-1: Pump Laser Specifications

	1 kHz	5kHz
Energy per pulse (mJ)	10	3.0
Average Power (W)	10	15
Wavelength (nm)	527	527
Beam Diameter (nominal)	6 mm	6 mm
Energy Stability (±p/p)	<2%	<3%
Beam Profile	Multi-mode, uniform intensity	
Polarization	Linear horizontal	

Versions of *Spitfire* amplifiers other than those listed in this manual may be pumped by other lasers. In addition, earlier *Spitfire* systems may be pumped by lasers such as the Spectra-Physics *Merlin.* Contact your Spectra-Physics representative for more information.

Modelocked Seed Laser

The *Spitfire* was designed with a *Tsunami* or *Mai Tai* mode-locked Ti:sapphire seed laser in mind. These are exceptionally stable systems. Spectra-Physics is not responsible for problems caused when a laser other than one of these is used to seed the *Spitfire* laser

The seed laser must meet the following specifications:

Table 5-2: Seed Laser Specifications

Wavelength	750−950 nm	
Power	> 400 mW	
Beam Diameter at 1/e^2 points	< 2 mm	
Stability	< 1% rms	
Pulse Length	< 85 fs < 60 fs < 30 fs < 1.3 ps	Spitfire F, Spitfire PM Spitfire USF Spitfire 50FS Spitfire P
Polarization	Linear vertical	
Beam Divergence, full angle	< 0.6 mrad	

Preparation

Location and Layout

Each user will probably have unique layout requirements based on the application and the requirements and layout of the experiment, so when choosing a layout, please consider the following:

- The *Spitfire* head covers an area of table space as follows:
  5 x 2 ft (1.9 x 0.75 m) for the *Spitfire 50 FS*
  4 x 2 ft (1.5 x 0.75 m) for the *Spitfire F, P, PM, USF*
- Allow sufficient space around the assembly for water hoses, high-voltage connections, etc.
- Select a location where the electrical utilities for all the laser systems are readily available. Spectra-Physics strongly recommends that the laser system be located in a laboratory environment, i.e., a room that is free from dust and drafts and does not exhibit any large temperature fluctuations. Room temperature should be maintained to within ± 2°C during operation.
- For stability, the entire system should be placed on a single, standard optical table.
- Because occasional adjustments might be required to optimize performance, position the *Spitfire* to allow easy access to its internal controls.
- Place the seed laser as close as possible to the *Spitfire* to avoid beam instability problems (such as those caused by unstable routing mirrors or by too many mirrors). Only use stable routing mirrors.
- Do not leave exposed any laser beam that travels more than 3 inches (7.5 cm).
- Both the pump laser and mode-locked seed laser must operate within the specifications listed earlier.

The *Spitfire* is shipped pre-assembled, but some optics have been removed and carefully wrapped for protection during shipment. Leave them wrapped at this time. The Spectra-Physics representative assigned to perform the initial installation will unwrap and install these optics.

Required Utilities

The *Spitfire* requires access to 110/120 Vac, 15 A, single-phase power. The seed and pump laser systems have electrical and cooling requirements as well. Before beginning installation, refer to the user manuals for those units.

Make sure proper service is available at the site before the Spectra-Physics field technician arrives for the initial installation.

Recommended Diagnostic Equipment

The following equipment is recommended for day-to-day operation of the *Spitfire*:

- a power meter capable of measuring between 10 mW and 20 W average power (e.g., Ophir, Scientec, Molectron) from 500 nm to 900 nm
- a fast CRT analog oscilloscope capable of 300 MHz or better (e.g., a Tektronix 2467, 7104 or 2465)
- a fast photodiode with a 2 ns rise time or better (e.g., an Electro-Optics Technology Model ET 2000)
- IR viewer and IR card
- an autocorrelator (e.g., a Spectra-Physics Model SSA)

In addition, the following equipment should also be available during installation, maintenance and/or troubleshooting:

- a small, low-divergence HeNe laser (for alignment)
- two broadband mirrors and mounts (for aligning the HeNe to the system)

Tools Required:

The following tools may be needed during installation, maintenance and/or troubleshooting:

- three gimbal mounts with 4–6 in. adjustable height
- alignment pins
- 10 in. (25 cm) scale
- three silver mirrors for the above mounts
- #1 Phillips screwdriver
- white business card
- trim pot screwdriver
- lens tissue
- standard U.S. hex ball-driver set
- English and metric scales (rulers)
- gel linear polarizing film

Interconnect Diagrams

The figures below are schematic representations of the main signal and control connections between components of the *Spitfire* 1 kHz (Figure 5-1) and 5 kHz systems (Figure 5-2). For clarity, the more obvious connections are not shown (ac power for the *SDG II* for example).

Also not shown are the power, water, and control connections for the pump laser and seed laser. Refer to the *Mai Tai* or *Tsunami* (seed laser) and the *Evolution* (pump laser) user's manuals for this information.

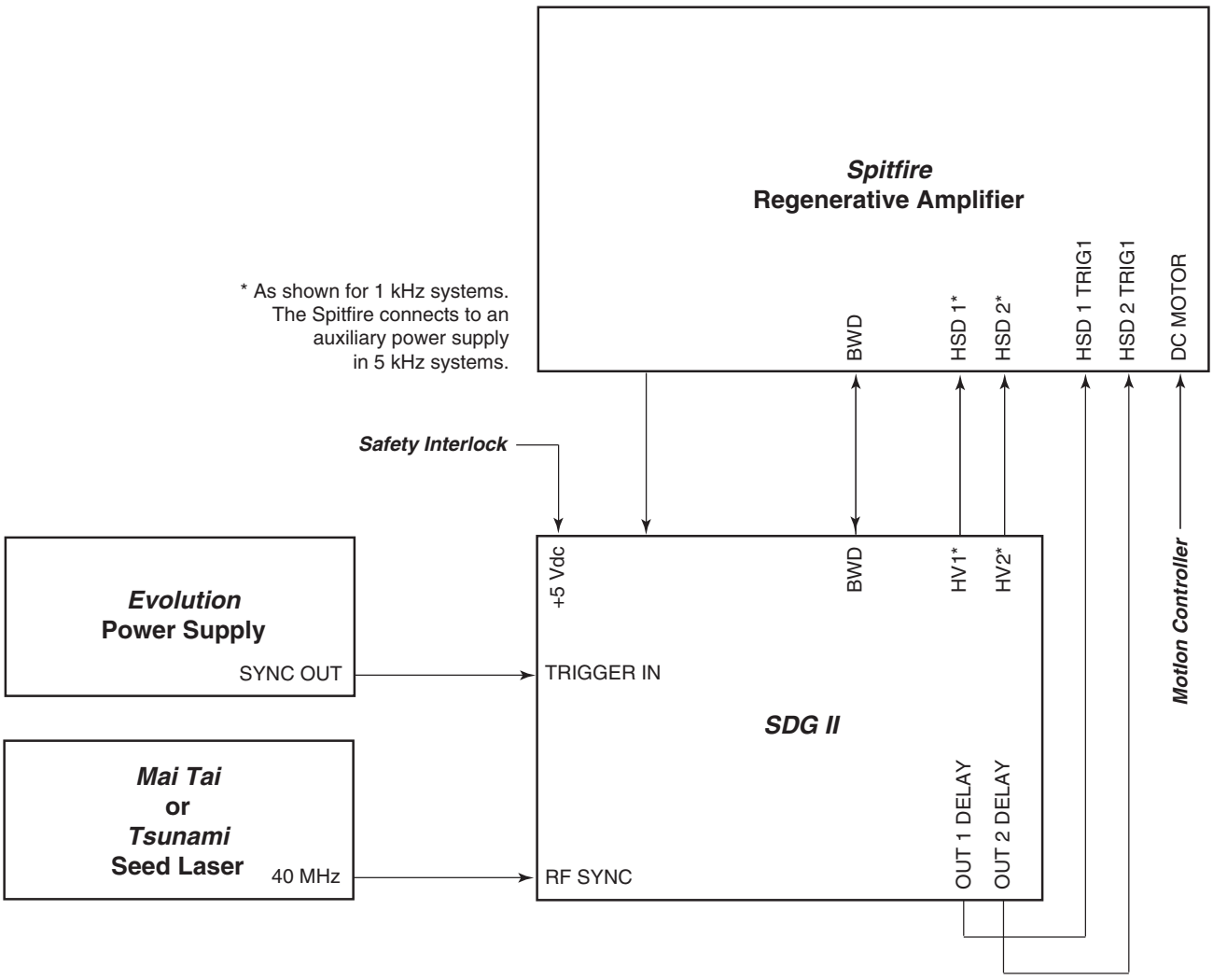


Figure 5-1: *Spitfire* Interconnect Diagram (1 kHz)

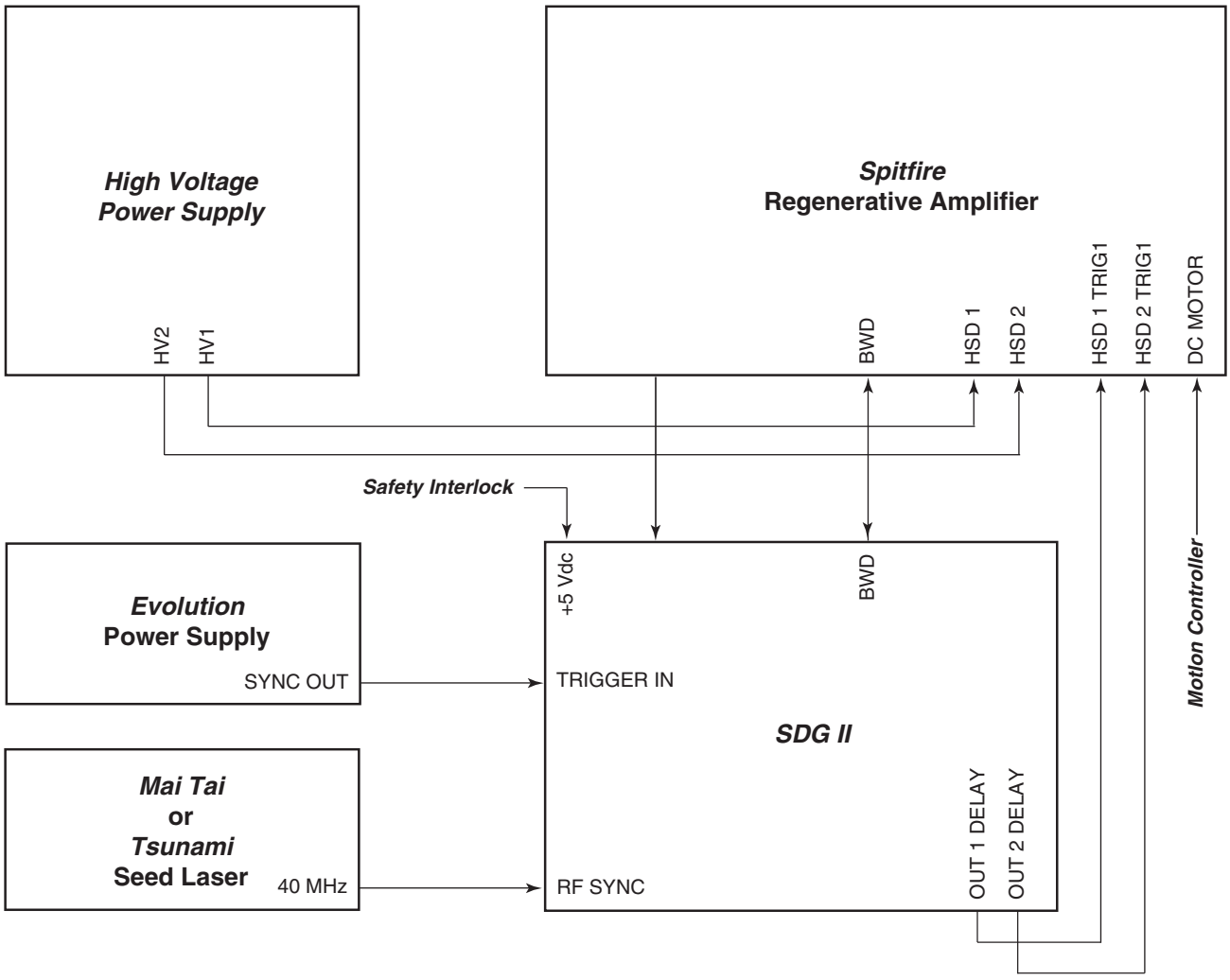


Figure 5-2: *Spitfire* Interconnect Diagram (5 kHz)

Chiller

The *Spitfire* Ti:sapphire amplifier rod must be cooled to avoid damage. The cooling water provided by the chiller for the *Mai Tai* or *Tsunami* laser is shared by the amplifier rod. This provides adequate thermal protection for the rod. The water flow to the amplifier is in series downstream from the *Mai Tai* or *Tsunami* as shown in Figure 5-3.

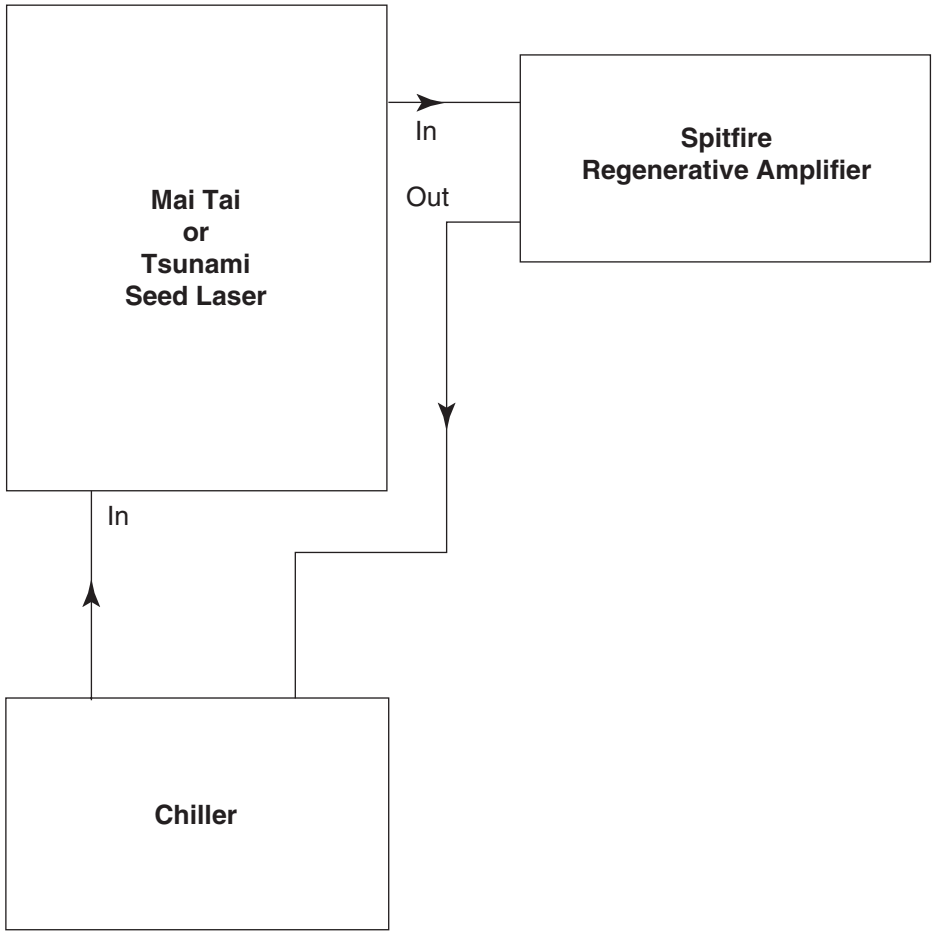


Figure 5-3: Serial Connections for Chiller Water

Chapter 6 Operation



Laser radiation is present. Safety glasses of OD 4 or greater at all lasing wavelengths must be worn at all times when operating this laser system.

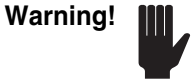
Refer to Appendix A for information about controlling the system via computer using the RS-232 interface on the *SDG II*.

It is recommended that the following equipment be kept on hand:

- a power meter capable of measuring between 10 mW and 20 W average power from 527 nm to 900 nm
- a fast photodiode with a 2 ns rise time or better
- a fast CRT analog oscilloscope capable of 300 MHz or better
- IR viewer and IR card
- an autocorrelator (e.g., Spectra-Physics SSA)

Start-up Procedure

Inspect the optic surfaces before the *Spitfire* is turned on and blow off any dust with dry nitrogen. Clean the optics as necessary.



Except for blowing off dust with dry nitrogen, *the gratings and the gold-coated mirror cannot be cleaned*. Attempting to clean these components will result in permanent damage.

1. Turn on the seed laser system (*Mai Tai* or *Tsunami*), including the chiller, as described in its user's manual.
2. Check the alignment of the mode-locked beam into the *Spitfire*. Optimize the alignment, if necessary, following the procedure below, "Seed Beam Alignment into the Regenerative Amplifier."
3. Turn on the *SDG II*. Push the reset button on the front panel for the BWD interlock. If the seed laser is properly mode-locked, both BWD LEDs will be lit. Refer to the procedures in Chapter 8, "Maintenance and Troubleshooting," if the LEDs indicate a problem (one or both are not on).
4. Enable OUT 1 DELAY and OUT 2 DELAY on the *SDG II*.
5. Turn on the pump laser (*Evolution*) as described in the user manual that accompanies it. Allow for the specified warm-up period.

6. Adjust the *Spitfire* repetition rate using the INPUT DIVIDE control on the *SDG II* (if so desired).

Optimizing Pulse Compression

Temperature changes or similar variations in the environment of the *Spitfire* may require adjusting the compressor to optimize the pulsed output.

Use the Motion Controller to set the compressor length for optimum compression. This adjustment is critical for femtosecond operation: the length must be within about 0.1 mm of the optimum. The best way to set the compressor length is to monitor the pulse width using an autocorrelator while using the Motion Controller to adjust the horizontal retroreflector.

The compressed pulse should look like that shown in Figure 6-1:

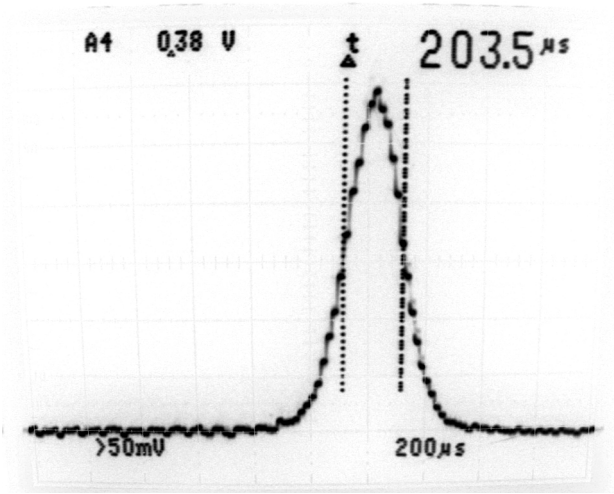


Figure 6-1: Autocorrelation of a Well Compressed Pulse

If you do not have access to an autocorrelator, optimize the pulse length by observing the output on a white business card. When the compressor length is correct, the beam on the card will appear blue in the center due to high peak power frequency doubling in the treated paper.

Shut-down Procedure

1. Before shutting down, enter *Spitfire* output power into a system log, along with the level of the pump laser and the timing parameters of the *SDG II*.
2. Disable OUT 1 DELAY and OUT 2 DELAY.
3. Power down the *SDG II*.
4. Turn off the pump laser (*Evolution)* as described in its user's manual. Note that the chiller must remain on if the *Evolution* power supply is left on.
5. Power down the seed laser (*Mai Tai* or *Tsunami)* as described in their user's manual. The chiller for the seed laser should always remain on.

Basic Performance Optimization

In addition to optimizing the optical length of the compressor as described in the previous section, the parameters that should be optimized are:

- stability of the seed pulses
- seed beam alignment into the regenerative amplifier
- beam uniformity
- build-up reduction time (optimizing the regenerative amplifier)

These parameters should not need to be checked or optimized on a daily basis; nevertheless they are fundamental to proper operation of the system and so are considered routine.

For convenience, Figure 6-2 shows the components used to align the seed beam into the *Spitfire* regenerative amplifier. The details of the optical design of the *Spitfire* models are described in Chapter 7.

Stability of the Seed Pulses

The mode-locked output of the seed laser must be optimized to ensure good stability of the amplifier. Refer to the seed laser user's manual. In particular, it is important that the duration of the seed pulse be not too long because the stretcher may not sufficiently reduce the peak power to avoid damage to the *Spitfire* optics. Use a scanning autocorrelator to monitor the seed pulse duration.

Seed Beam Alignment into the Regenerative Amplifier

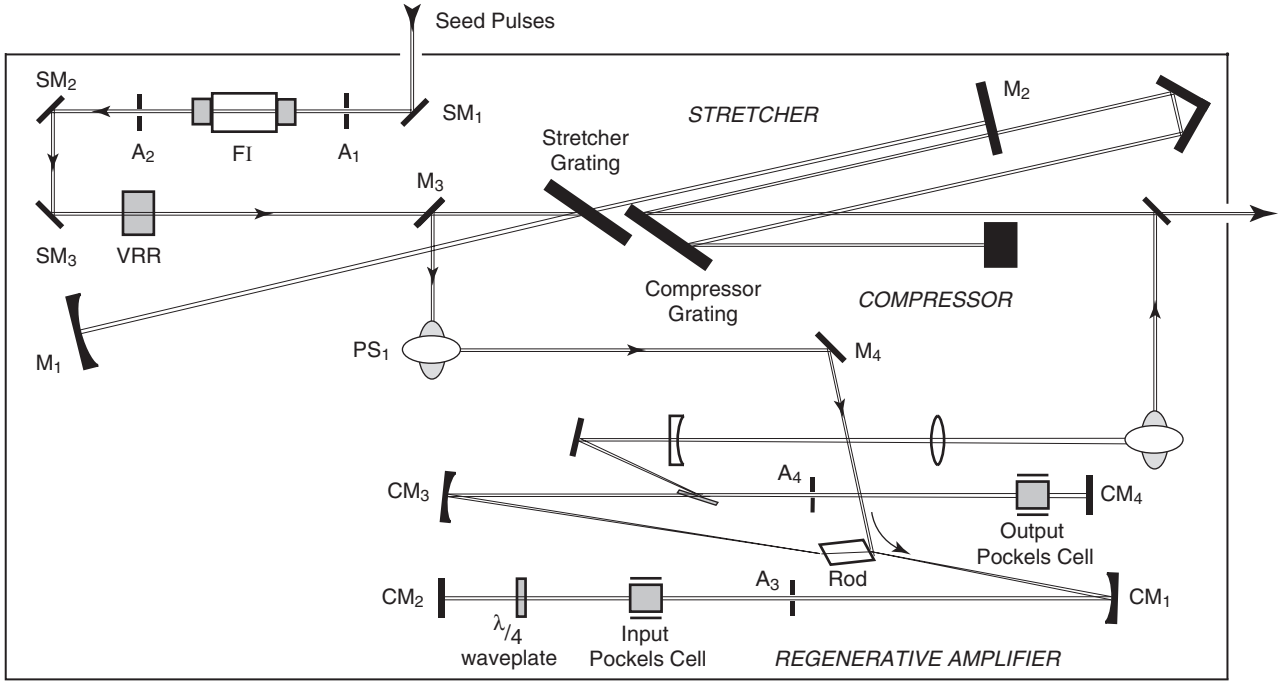


Figure 6-2: Optical Path for Seed Beam Alignment

The input beam is directed by SM_1 through the Faraday Isolator FI and the first two alignment apertures A_1 and A_2. It is then routed by seed mirrors SM_2 and SM_3 through the vertical retro-reflector VRR and onto the stretcher diffraction grating. SM_1, SM_2 and SM_3 all have vertical and horizontal adjustments.

1. Verify the pump beam shuttered is closed.
2. Check the alignment of the seed beam through apertures A_1 and A_2, and make small adjustments as necessary.
3. Rotate the gratings out of the beam path and use the IR card to verify the beam is well aligned through the apertures (not shown) at the entrance to the stretcher and the exit from the compressor.
4. Rotate the gratings to their original positions to resume normal operation. The 1st order diffracted beam should strike the center of gold mirror M_1.
5. Check the alignment of the seed beam into the amplifier. It is possible that the seed beam will have drifted slightly since the *Spitfire* was last operated. The beam should be aligned using mirrors M_3 and M_4 so that it is centered on the input Pockels cell and then onto cavity mirror CM_2.
6. Enough of the beam should pass back through the input Pockels cell, the Ti:sapphire rod, and the other components in the optical path so that it is visible on the IR card in front of cavity mirror CM_4. Use the IR card to make slight adjustments to mirror M_4, *not the cavity mirrors*, until you see a beam at CM_4.

Beam Uniformity

The *Spitfire* amplifier is designed to produce a near Gaussian output beam. Beam uniformity is best checked by visually inspecting burn patterns made on Eastman Kodak's Linagraph paper, commonly called “burn paper.” The beam can be incident on either side of the burn paper, giving different and often complementary information.

Warning!

When making burn patterns, keep the sample of burn paper in a transparent plastic bag in order to avoid getting residue on the optical surfaces. Be careful to avoid reflections from the plastic!

A poor beam is an indication of optical damage or misalignment, particularly the alignment of the pump beam. Refer to Appendix C for procedures to optimize pump beam alignment (refer to Chapter 7 for a description of the pump beam path.)

Caution!

Make only small and reversible changes to the pump beam alignment. The pump beam is tightly focused in the Ti:sapphire rod in the amplifier, and is easy to misalign. Refer to Appendix C for a complete description of the pump beam alignment procedure. It is recommended that you contact your Spectra-Physics representative before making adjustments to the pump laser.

Optimizing the Regenerative Amplifier

Use this procedure to minimize the time it takes for the amplified pulse to reach its peak level. This "build-up time" is compared to the time it takes for the *Spitfire* to amplify its own spontaneous emission in the absence of a seed pulse. Minimizing this relative time for the pulse to be amplified is called "build-up time reduction."

Note

Minimizing the build-up time reduction is fundamental to optimizing the performance of the *Spitfire*.

Danger!
Laser Radiation

In the following procedure, the regenerative amplifier is initially operated as an optically-pumped, Q-switched laser. In this configuration, the *Spitfire* is capable of producing >1.5 W of average power at a wavelength near 800 nm. Use appropriate caution.

1. Block the seed beam.
2. Open the pump laser shutter.
3. Disable OUT 2 DELAY. The regenerative amplifier will begin to operate as a laser. Allow it to stabilize for about 5 minutes.
4. Monitor the intracavity pulse using the output of the photodiode behind CM_4. Use a fast oscilloscope with a micro-channel plate screen, or a digitizing oscilloscope with a sampling rate greater than 2 GHz.

   Trigger the oscilloscope externally with the *SDG II* SYNC OUT DELAY. Set the time-base to 100 or 200 ns/div. Use a 50 Ω input impedance for the photodiode.

   The pulse should appear as shown in Figure 6-3.

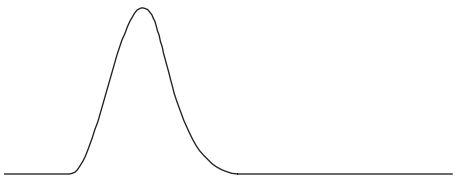


Figure 6-3: Appearance of Q-switched Pulse

5. Unblock the seed beam. The energy of the seed laser pulses will now overcome the energy of the spontaneous emission in the Ti:sapphire rod in the regenerative amplifier so that it now amplifies the seed pulses.

   The intracavity radiation should now look like that shown in Figure 6-4.

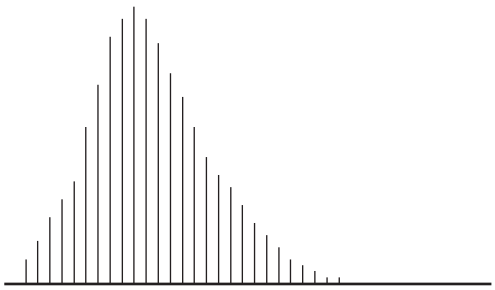


Figure 6-4: Intracavity Pulse Train

6. *Reduce the pulse build-up time to the minimum possible.* While monitoring the pulse train, make small, iterative adjustments to turning mirrors M_3 and M_4 (which direct the seed pulse into the resonator) so that the pulse train moves to the left on the oscilloscope screen.
7. Next, enable and adjust the *SDG II* OUT 2 DELAY until the intracavity pulse train looks like that shown in Figure 6-5. Note that the most stable performance is obtained by adjusting the timing so that the pulse train includes the one pulse that is just past the maximum.

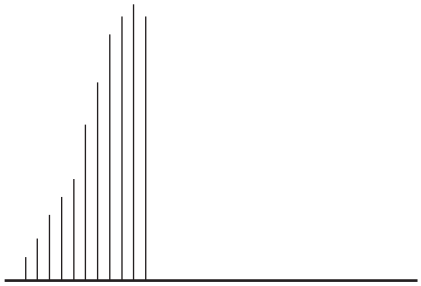


Figure 6-5: Intracavity Pulse Train with the Timing Set Correctly

8. Set up a fast photodiode to sample the output of the *Spitfire* and view the output pulse on the oscilloscope. (Use the same settings given in Step 4). A single, stable, output pulse should be displayed.
9. Observe the output mode. If adjustment is necessary, refer to the pump beam alignment procedure in Appendix C.
10. If the pulse amplitude is stable but there is evidence of a secondary pulse, make a slight adjustment to the OUT 2 DELAY control. If this does not produce a single, stable, cavity-dumped pulse, adjust OUT 1 DELAY by ±10 ns.

If this procedure does not produce a single, stable, cavity-dumped pulse, re-check and adjust the intracavity Q-switched pulse (i.e., block the seed laser beam again).

If a single, stable pulse is still not produced, contact your Spectra-Physics service engineer.

Re-Optimization

A change in room temperature or similar environmental factors may make re-optimization necessary. To do this:

1. Disable OUT 2 DELAY on the *SDG II*, and monitor the intracavity pulse as it builds up. It should look like that shown in Figure 6-4.
2. Block the seed beam into the *Spitfire,* and observe the intracavity Q-switched pulse (Figure 6-3) as it builds up.
3. If the Q-switched pulse is unstable in amplitude or time, make slight adjustments to end mirrors CM_1 and CM_4. With the oscilloscope triggered by the SYNC OUT DELAY on the *SDG II*, the Q-switched buildup time reduction should be approximately 100–150 ns.
4. Unblock the seed beam into the *Spitfire*. The pulse train should look like that shown in Figure 6-4. Make small, iterative adjustments to mirrors M_3 and M_4 to reduce the pulse buildup time as much as possible; that is, adjust it so that the pulse train shifts from right to left on the oscilloscope screen.

   By alternatively blocking and unblocking the seed beam into the *Spitfire*, the difference between the unseeded Q-switched time and the seeded pulse train time can be measured. This difference in buildup time should be approximately 50–80 ns.
5. Re-enable OUT 2 DELAY on the *SDG II*. Again, the intracavity pulse train should look like that shown in Figure 6-5. If it does not, adjust OUT 2 DELAY so that the highest amplitude pulses in the train remain.
6. Position the photodiode at the output port of the *Spitfire* to look at the ejected pulse. Make slight adjustments to OUT 2 DELAY to eject the pulse that has the best stability. Adjust OUT 1 DELAY slightly if there is evidence of a secondary pulse being ejected.

Chapter 7 The Spitfire Beam Path

On occasion, it might be necessary to make adjustments to the *Spitfire* internal optical components. The beam path and its adjustment through the *Spitfire* are described below.

Note

When describing the beam path, "left" and "right" refer to the direction of travel moving along the beam from input to output.

A few of the more complex optical elements require some initial description. Refer to Figure 7-1 below.

Faraday Isolator—protects the seed laser components by absorbing any reflected power that is generated in the amplifier and absorbing pulses that are not selected for amplification. There are no adjustments on this device.

Tall Stretcher End Mirror (M_2)—is about 95% reflective so that only about 5% of the beam passes through it and is detected by the bandwidth detector (BWD) located behind the mirror. The end mirror reflects the beam back onto the gold Mirror (M_1). Both M_1 and M_2 have vertical and horizontal adjustments.

Bandwidth Detector (BWD) (not shown in the drawings)—is a safety device that protects the system when there is not enough bandwidth in the seed pulse for it to be properly spread by the stretcher (usually caused by a misaligned seed laser or one with poor mode-locking). See Chapter 4, "Controls, Indicators and Connections," for more information about the BWD. This device is pre-set at the factory.

Vertical Retroreflectors (x2)—comprise a pair of flat mirrors at right angles that translate the beam up or down and reflects it back on a parallel path. There is one of these assemblies in the stretcher and one in the compressor; each has vertical and horizontal adjustments.

Horizontal Retroreflector—translates the beam sideways and reflects it back on a parallel path. This compressor assembly has vertical and horizontal adjustments. In addition, the horizontal retroreflector is mounted on a translational track that has a dc motor and motion controller.

Polarizer—is an optical element that, as used in the *Spitfire*, is transparent to horizontally polarized light and reflects (rejects) vertically polarized light. It is used to direct amplified pulses into the compressor. Mounting screws provide vertical and lateral movement for alignment. There are no other adjustments on this device.

Stretcher and Compressor Beam Paths

Although the beam passes from the stretcher into the amplifier and then to the compressor, *the beam path in the stretcher and in the compressor are described together first* since they share the same compartment and their treatment of the beam is similar.

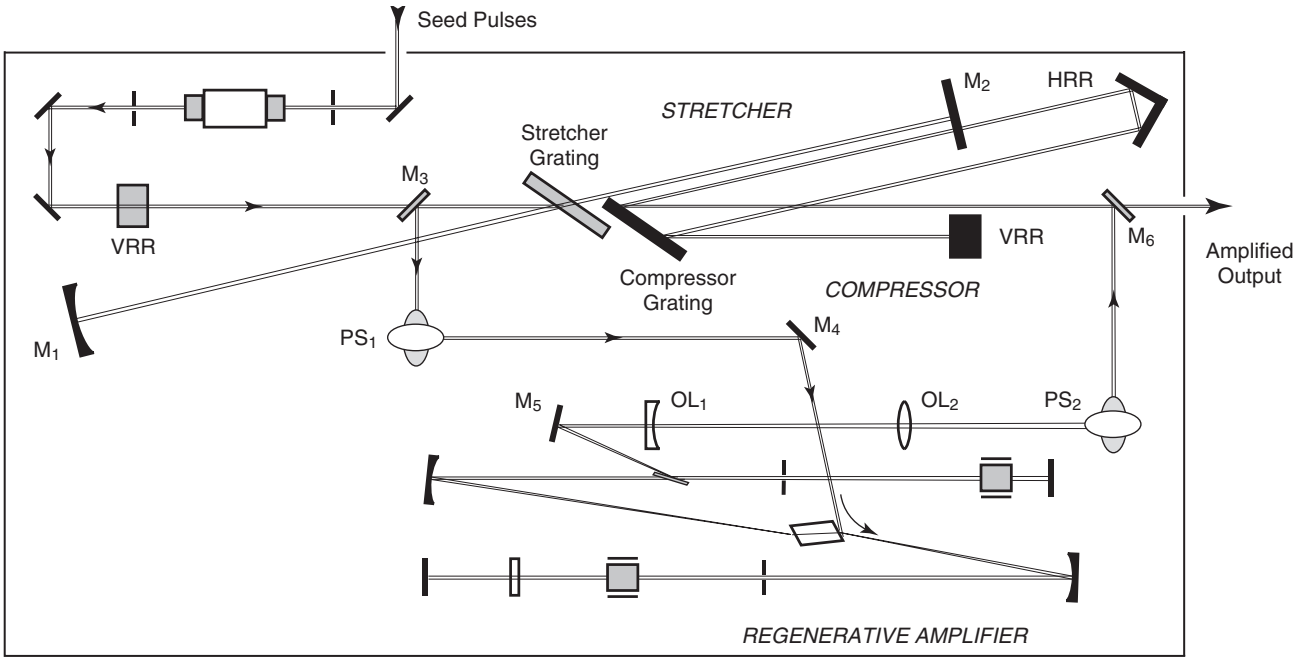


Figure 7-1: Optical Components in the *Spitfire F* (Stretcher and Compressor)

The different versions of the *Spitfire* have different stretcher and compressor designs as a result of their different pulse widths. Basically, each version uses its own set of gratings that are set at different angles to the beam. There are also other relatively minor differences.

Although your version of the *Spitfire* may differ slightly, study the beam path through the *Spitfire F* model. Differences in the design of the stretcher/compressor in the other versions are described relative to the *Spitfire F* in later sections.

After passing through the Faraday isolator, the seed laser beam is horizontally polarized and remains so in the stretcher. The polarization of the beam is changed in the amplifier, but when it enters the compressor, it is again horizontally polarized.

Numbers are used in the following drawings to track the path of the beam as it passes from optic to optic. The numbers are not used either to name the optic itself or to indicate the position of the beam. Refer to Figure 7-1 for the abbreviations used to name components in the stretcher and compressor.

Note

Shorter and longer wavelengths strike the optical components in the stretcher and compressor at different locations. Figure 7-2 shows how the wavelengths are separated.

The Spitfire F Stretcher

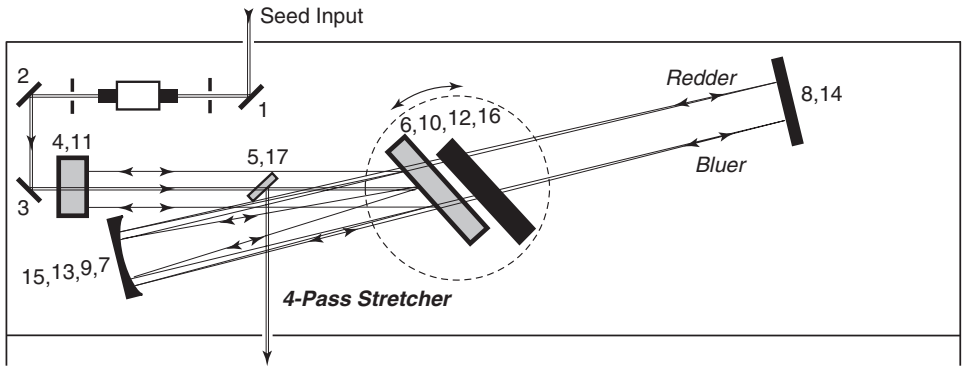


Figure 7-2: *Spitfire F* Stretcher Beam Path

The vertically polarized seed beam is first routed through the Faraday isolator (1,2,3) before entering the stretcher.

1. To enter the stretcher, the seed beam passes *through* a gap in the vertical retroreflector VRR (4) and *over* the pick-off mirror, M_3 (5).
2. The grating spreads the beam spectrally (6) and directs the broadening beam onto the center of the concave gold mirror, M_1 (7). The grating mount has a single adjustment that rotates the grating to change the angle of incidence of the beam. The stretcher grating shares its mount with the compressor grating.
3. M_1 is angled slightly upward to reflect the beam *over* the grating (8) onto the tall stretcher end mirror, M_2. The concave gold mirror and the tall stretcher end mirror have vertical and horizontal adjustments.
4. M2 reflects the beam back *over* the grating to M_1 (9), which returns it to the grating (10).
5. The grating reflects the collimated beam toward the bottom of the vertical retroreflector VRR (11). Notice that the path of the redder wavelengths is longer than that of the bluer wavelengths and, therefore, lags behind the bluer wavelengths.
6. The beam now retraces its path back through the stretcher. VRR (12) reflects the beam back to the top of the grating. The spectrum is temporally spread even further as the redder wavelengths again take the longer path. Passing the beam through the stretcher one more time (13, 14, 15, 16), it is focused back into a round beam. However, the bluer components are now well ahead of the red.
7. Because the beam hits high on the concave mirror, it is reflected to the bottom of the grating, and as it leaves the grating it is now low enough to be picked off by mirror M_3 (17). It exits the stretcher and is routed into the regenerative amplifier. M_3 has vertical and horizontal adjustments.

The Spitfire F Compressor

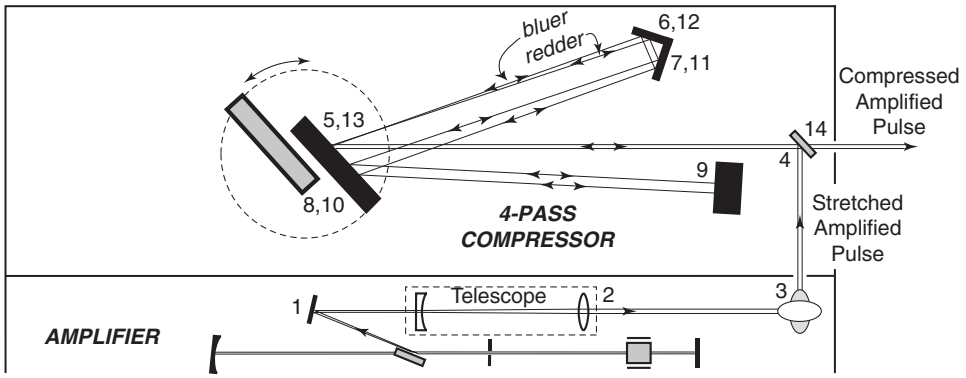


Figure 7-3: *Spitfire F* Compressor Beam Path

The temporally stretched, pulsed beam passes from the stretcher into the regenerative amplifier. After it achieves its maximum level of amplification, the beam is then ejected out of the regenerative amplifier by the horizontal polarizer (see Figure 7-6). The mechanism for ejecting the beam from the amplifier is discussed in "The Ti:Sapphire Regenerative Amplifier" on page 7-7.

1. M_5 (1) directs the vertically polarized beam through the expanding telescope (2) (comprising OL_1 and OL_2) to reduce the beam intensity in the compressor.
2. Polarizing periscope PS_2 (3) rotates the beam to horizontal polarization and directs it to the compressor routing mirror M_6 (4), which then sends it onto the right side of the compressor grating (5). M_6 has vertical and horizontal adjustments.
3. The grating spreads and reflects the beam towards the horizontal retroreflector HRR (6, 7), with the redder wavelengths on the right and the bluer wavelengths on the left. The grating mount has a rotational adjustment which it shares with the stretcher grating.
4. The horizontal HHR steps the beam over about two inches, flips the ends of the spectrum, and returns the beam to the lower left side of the grating (8). The redder wavelengths now take the shorter path.
5. The beam is reflected by the grating and impinges on the VRR (9) where it is stepped upwards an inch and is sent back to the top left side of the grating (10), which begins to refocus the beam and reflects it to the horizontal retroreflector (11, 12).
6. The horizontal retroreflector flips the beam around again and sends it back to the grating (13) where the beam is compressed back close to its original duration.
7. The beam is reflected *over* M_6 (14) and exits the *Spitfire*.

Spitfire USF Stretcher and Compressor

The layout of the *Spitfire USF* is identical to that of the *Spitfire F*. The only difference is that the grating ruling density is reduced in the *Spitfire USF* to accommodate its shorter pulses (and, therefore, broader spectral bandwidth). The gratings are also set at a different angle to the beam.

Spitfire P Stretcher and Compressor

The narrower spectrum of picosecond pulses (as compared to femtosecond pulses) requires greater dispersive power from the gratings in order to adequately reduce the peak power of these pulses before they can be safely amplified. Therefore, the *Spitfire P* uses gratings with an increased ruling density, compared to that used on the *Spitfire F,* and set at a different angle to the beam.

In addition, the stretcher and compressor incorporate additional folding mirrors to increase the length of the beam path and give the separated wavelengths adequate distance to separate and recombine spatially. Figure 7-4 illustrates the stretcher and compressor beam path for picosecond operation.

Note

For clarity, the dispersion of the beam is not shown in the same detail in Figure 7-4 as in the other stretcher/compressor figures.

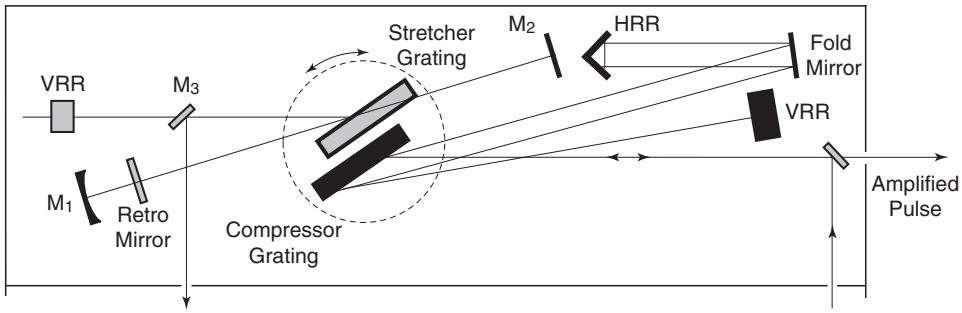


Figure 7-4: Modifications for the *Spitfire P*

Spitfire PM Stretcher and Compressor

The picomask version of the *Spitfire* is identical to the *Spitfire F*, except that it uses the same gratings as the picosecond amplifier, and a special mask aperture is added to the stretcher cavity to reduce the bandwidth of the femtosecond seed pulses. The *Spitfire PM* does not require the extra path length in the stretcher and compressor that is needed by the *Spitfire P.*

This mask and its position in the stretcher in front of M_1 are shown in Appendix B, “Changing To and From PicoMask Operation,” which provides instructions for converting *Spitfire F* and *Spitfire USF* femtosecond models to and from picomask operation.

Spitfire 50FS Stretcher and Compressor

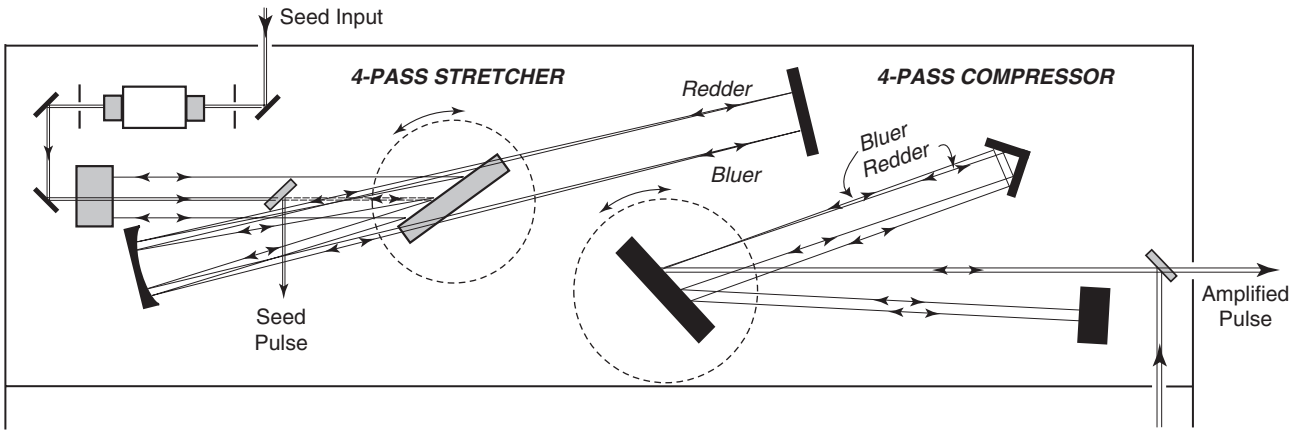


Figure 7-5: *Spitfire 50FS* Stretcher and Compressor Beam Path

Amplification of pulses of the shortest duration requires that extra attention be paid to correcting the dispersion that occurs within the amplifier cavity.

Each Fourier component frequency of a pulse experiences a slightly different index of refraction as it propagates though a material, causing a time delay between the different frequencies. Group Velocity Dispersion (GVD) is defined as the variation in the time delay as a function of wavelength. Typically, GVD causes red frequencies to travel faster than blue frequencies. The effect is more pronounced for shorter pulses, such as those amplified by the *Spitfire 50FS*.

In addition to GVD, the pulse width is affected by the nonlinear index of Ti:sapphire, which results in self phase modulation (SPM). As the pulse propagates through the Ti:sapphire material, the leading edge is "red-shifted" by an increasing index of refraction. Conversely, the trailing edge of the pulse is "blue-shifted." (More information about GVD, SPM, and dispersion compensation can be found in the *Mai Tai* or *Tsunami* user's manuals.)

In order to achieve near transform-limited output pulses, it is necessary to compensate for the pulse spreading caused by positive GVD and SPM. This is accomplished by using a compressor grating with a higher ruling density than the stretcher grating.

Such a design no longer permits the same ruling density to be used for both the stretcher and the compressor grating as in the other *Spitfire* models. Furthermore, using a different ruling density for each gratings requires each grating to be presented to the beam at a different angle, which then varies as the unit is tuned for wavelength. Therefore, the *Spitfire 50FS* uses separate grating mounts, rather than the shared, single adjustment mount found in the other *Spitfire* models.

However, the same four-pass design can be used for the beam paths in the stretcher and in the compressor in order to obtain adequate spatial separation of the pulse's wavelength components. The beam in the *Spitfire 50FS* follows the same sequence from optic to optic as outlined earlier for the *Spitfire F*.

The Ti:Sapphire Regenerative Amplifier

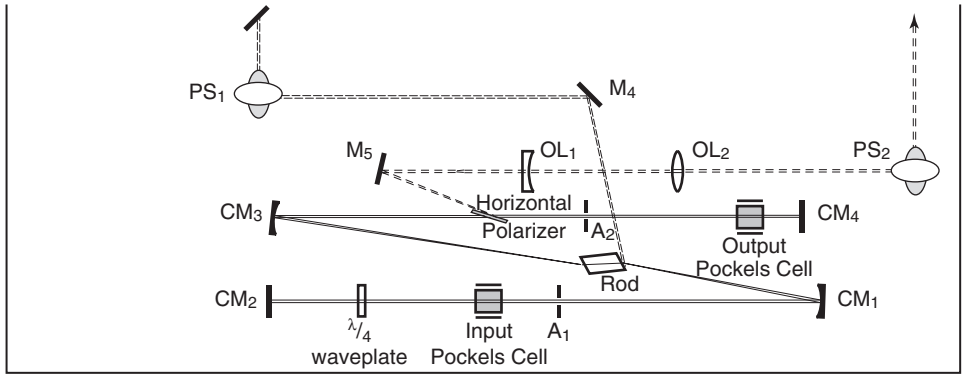


Figure 7-6: Regenerative Amplifier Optical Components

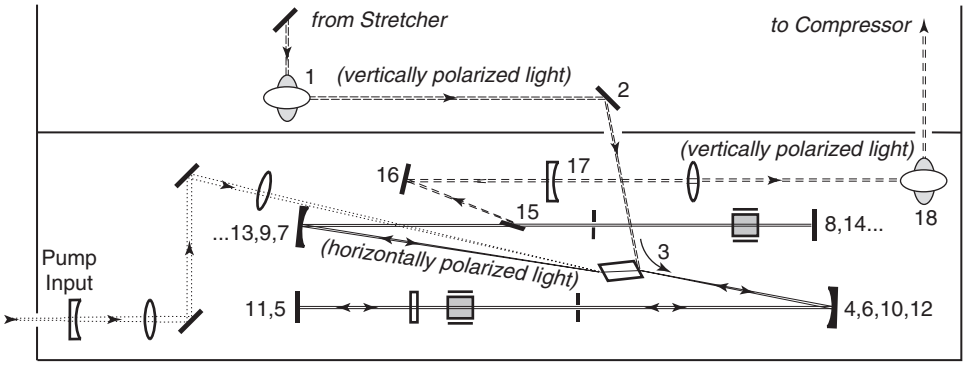


Figure 7-7: Regenerative Amplifier Beam Path

Spitfire models all make use of a regenerative amplifier in a Z-shaped folded cavity. Refer to Figure 7-6 for component names ("M_1", "$\lambda/4$" etc.). The beam path is shown in Figure 7-7. For clarity, some components, such as apertures and lenses, are not shown.

The Seed Beam Path

1. Horizontally polarized pulses from the stretcher are rotated (1) to vertical polarization by the polarization rotating periscope PS_1 and are directed into the amplifier cavity by mirror M_4 (2).
2. The Ti:sapphire rod, cut at Brewster's angle for horizontally polarized light, reflects the vertically polarized pulses off its surface (3) and directs them to the first cavity mirror CM_1 (4), which directs the pulses to the input Pockels cell.

At this point, the pulsed beam is in the amplifier cavity.

Whether a particular pulse remains in the cavity to be amplified is determined by the input Pockels cell. When this Pockels cell is off, it is transparent to both vertically polarized and horizontally polarized light. When the input Pockels cell is on, combined with the ¼ waveplate ($\lambda/4$), it rotates the polarization of the beam 90°.

One of three things will now happen:

Case (a)—the input Pockels cell is off when the pulse arrives, the pulse passes through the cell and is reflected back to the same Pockels cell. If this Pockels cell is still off when the pulse returns, the pulse is rejected after one round trip through the amplifier.

Case (b)—the input Pockels cell is on when the pulse arrives, the pulse is not selected and is rejected without passing through the Ti:sapphire rod.

Case (c)—the input Pockels cell is off when the pulse arrives but is turned on after the pulse travels through it and before the pulse is reflected back to the same cell. The pulse is now selected. In this case, the pulse makes about 20 round trips in the cavity, gaining in amplitude with each pass, and is released into the compressor by the activation of the output Pockels cell.

Note



As long as the selected pulse remains horizontally polarized, it remains in the cavity. Whenever a pulse arrives at the Ti:sapphire crystal as vertically polarized, it is reflected off the surface and is not amplified.

Pulse selection is accomplished by using the polarization rotating properties of the passive $\lambda/4$ together with the input Pockels cell. Pulses at kilohertz rates are selected for amplification while the remaining megahertz seed pulses are rejected. Control of pulse selection is determined by the *SDG II*, as described in Chapter 4.

Each of these three cases is now described in detail:

Case (a): the input Pockels cell is off and stays off (pulse is rejected)

1. The vertically polarized pulse reflects off CM_1 (4), passes through aperture A_1, through the inactive input Pockels cell, and is rotated 45° as it passes through $\lambda/4$. It reflects off CM_2 (5) and rotates another 45° as it passes through $\lambda/4$ again.
2. The pulse, now horizontally polarized, passes through the inactive input Pockels cell again, through the aperture, and is reflected by CM_1 (6), this time to the Ti:sapphire rod. Because it is now horizontally polarized, it passes through the rod and picks up first-pass gain.
3. The pulse is reflected by CM_3 (7) through the horizontal polarizer, through aperture A_2 and through the inactive output Pockels cell.
4. The pulse reflects off CM_4 (8) and passes back through the inactive output Pockels cell, through aperture A_2, through the horizontal polarizer, reflects off CM_3 (9), and passes back through the Ti:sapphire rod for second pass gain.
5. The beam reflects from CM_1 (10), passes through the inactive input Pockels cell, and is again reflected back from CM_2 (11). Having passed twice through $\lambda/4$, it is now vertically polarized and is reflected from the surface of the Ti:sapphire rod and out of the amplifier cavity to M_4. Once rejected, the pulse passes back through the stretcher and is absorbed by the Faraday isolator.

Case (b): Input Pockels cell is already on (pulse is rejected)

1. The incoming vertically polarized pulse reflects off CM_1 (4) and passes through aperture A_1. But this time, as it passes through the now active input Pockels cell, it is rotated 45° by the cell and another 45° as it passes through $\lambda/4$, becoming horizontally polarized. After reflecting off CM_2 (5), it is rotated another 45° by $\lambda/4$ and another 45° by the still active input Pockels cell, and returns to vertical polarization. It passes through aperture A_1, reflects off CM_1 (6) and is reflected off the Ti:sapphire surface and ejected out of the cavity.

Case (c): Input Pockels cell is off and then is turned on (pulse is selected)

1. The soon-to-be-selected, vertically polarized pulse reflects off CM_1 (4), passes through aperture A_1, through the inactive input Pockels cell, and is rotated 45° as it passes through $\lambda/4$. It reflects off CM_2 (5) and rotates another 45° as it passes through $\lambda/4$ again.

   This time, after the pulse passes back through the inactive Pockels cell and travels toward CM_1 (6), the input Pockels cell is turned on. (The output Pockels cell remains off for now.)

   The pulse remains in the cavity because it remains horizontally polarized. (Since the input Pockels cell is on, the pulse is flipped 180° each time it traverses the input path, leaving its polarization unchanged.) The pulse is then amplified each time it passes through the crystal. The pulses that follow behind the selected pulse arrive with the input Pockels cell already turned on. These following pulses remain vertically polarized as in Case (b), and are discarded.
2. After the selected pulse has passed through the crystal about 20–25 times (6 through 14...), it has reached its optimum amplification. The output Pockels cell is now turned on just before the pulse returns to it (the precise timing is set by the *SDG II*), and the pulse now finds the output Pockels cell acting as a ¼ waveplate. It is rotated 45° going in and 45° reflecting back from CM_4 and becomes vertically polarized. It is reflected out of the cavity by the horizontal polarizer.
3. The vertically polarized, amplified pulse is reflected by the polarizer (15) to mirror M_5 (16).
4. To protect the compressor optics, the beam is expanded by a telescope (17) (comprising negative and positive lenses OL_2 and OL_3) to reduce pulse power density.
5. The expanded beam is directed into the compressor by the polarization rotating periscope PS_2 (18), which changes the vertically polarized light from the amplifier to horizontally polarized light.
6. M6 (see Figure 7-1) directs the beam into the compressor chamber.

The Pump Beam Path

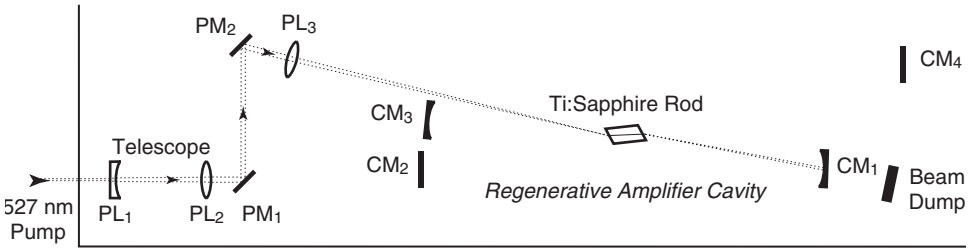


Figure 7-8: Pump Beam Path

The pump beam path is controlled as follows:

A telescope comprising negative lens PL_1 and positive lens PL_2 enlarge the pulsed beam from the pump laser. (The beam is enlarged so that it can be better focused into the Ti:sapphire rod.) The lens mounts have vertical and horizontal adjustments.

Pump mirror PM_1 directs the enlarged beam from the telescope to PM_2. PM_1 and PM_2 are also used to set the height of the pump beam so that it is centered on pump lens PL_3. Pump mirror PM_1 has vertical and horizontal adjustments.

PL_3 focuses the pump beam in the Ti:sapphire rod. The lens mount has vertical and horizontal adjustments as well as movement in the direction of the beam to focus it.

The 527 nm (green) light from the Nd:YLF pump laser is horizontally polarized, allowing it to enter and be absorbed by the Ti:sapphire rod. Roughly 80% of the pump beam is absorbed by the rod.

The fraction of the high-power pump beam that is not absorbed by the Ti:sapphire rod transmits through the rod onto CM_1. CM_1 does not reflect a significant amount of 527 nm light, and, instead, allows it to pass through where it is absorbed by the beam dump behind it.

Chapter 8 Maintenance and Troubleshooting



Exceptional care must be taken when operating the *Spitfire* with the covers removed. Laser protective eyewear *must* be worn to protect the eyes from all wavelength emissions.

Try This First

If the *Spitfire* is producing pulses but performance has degraded, first verify the following:

- the BWD interlock is properly set
- the seed laser is operating properly and its shutter is open
- the pump laser is operating properly and its shutter is open
- the pump beam routing mirror PM_2 is properly set

If the components above are operating properly, you may only need to adjust the pump power or pump beam routing mirror PM_2 to optimize output. Refer to Appendix C before attempting these adjustments.

If the *Spitfire* is not producing amplified pulses, first verify the following:

- there is sufficient pump power
- the pump beam has not become misaligned
- the *SDG II* OUT 2 DELAY is sufficiently greater than OUT 1 DELAY
- the +5VDC ENABLE switch on the *SDG II* back panel is in the disabled or down position.
- the Pockels cells are properly connected, triggered and operating
- the intracavity apertures are not blocking the beam

If these criteria are all met, inspect the internal optics for cleanliness and damage. Use the procedure below to clean optics as needed. Heed the warnings regarding cleaning! Not all optical surfaces can be cleaned, other than by blowing dust off with dry nitrogen.

If the troubleshooting and corrective procedures in this chapter do not solve the problem, please contact your Spectra-Physics representative before taking further action. Contact information is included in “Customer Service” on page 8-6.

Cleaning Optics

The *Spitfire* has been designed for minimal maintenance. However, from time to time, depending on the laboratory environment, it may be necessary to clean the optics. The following materials are required:

- Reagent grade methanol or acetone
- Lens tissues
- Hemostat (surgical pliers)
- Eyedropper

Warning! 

Do not attempt to clean the surfaces of the gratings and the gold-coated mirrors. These optical surfaces can only be blown clean with dry nitrogen. Attempting to clean these components will permanently damage them.

The optics in the *Spitfire* should be carefully cleaned with soft optical tissue and reagent grade methanol or acetone as described below.

1. Always wash your hands first.
2. Wear finger cots whenever optics are handled.
3. Hold one sheet of lens tissue over the optic to be cleaned.
4. Using the eyedropper, place a single drop of good quality methanol on top of the lens tissue.
5. Drag the lens tissue across the optic only once.
6. If a residue of solvent is left on the optic, repeat the procedure using less solvent and a new lens tissue until no residue remains.

For hard to reach optics:

1. Wear finger cots or gloves.
2. Fold a piece of lens tissue repeatedly to form a pad of approximately 1 cm wide.
3. Hold the pad with a pair of hemostats so about 3 mm of the folded edge protrudes from the hemostat blades.
4. Saturate the pad with methanol or acetone and shake dry.
5. Reach slightly on one edge of mirrors and wipe the surface of the mirrors toward the outside *in one motion*. *Use each pad only once!* Be very careful that the tip of the hemostats does not scratch the mirror.

Troubleshooting

Symptom: No Spitfire output

Possible Cause:	Corrective Action:
BWD interlock is open	Check the two BWD LEDs on the SDG II. If both LEDs are on, reset the interlock button of the BWD. If one or both LEDs are off, verify the seed laser is mode-locked and that the wavelength is centered as specified. Reset the BWD interlock button after restoring seed laser operation.
Seed laser is not functioning correctly	Refer to seed laser user's manual for further instructions.
Pump laser is not functioning correctly	Refer to the pump laser user's manual for further instructions.
Seed laser beam is misaligned	Optimize the seed laser alignment.
SDG II controls are disabled	Verify the unit is turned on and that the settings for OUT 1 DELAY and OUT 2 DELAY, SYNC ENABLE, and MODE control for CONTINUOUS or SINGLE SHOT operation are properly set.
Problem with high speed driver(s)	Contact your Spectra-Physics representative

Symptom: Regenerative Amplifier power is below specification

Possible Cause:	Corrective Action:
Optics are dusty	Use dry nitrogen to blow dust from the optics, with particular attention to the pump path and regenerative amplifier optics.
Optics are damaged	Check the optical components in the regenerative amplifier. If an optic has been damaged, contact your Spectra-Physics representative to arrange to have the optic changed. It may be possible to use an undamaged portion of the optic face and realign the regenerative amplifier as a temporary solution.
Seed laser beam is misaligned	Optimize the seed laser beam alignment.
Pump laser is power low	Optimize the pump laser power according to its user's manual.
Pump laser beam is misaligned	Optimize the alignment of the pump laser beam.
Regenerative Amplifier is misaligned	Refer to Appendix C.
Timing of Pockels cells is incorrect	Check the settings for OUT 1 DELAY and OUT 2 DELAY on the SDG II.

Symptom: Pulse has broadened out of specification

Possible Cause:	Corrective Action:
Compressor delay not be optimized	Adjust the compressor motor controller to get the shortest pulse.
Seed laser pulses are broadened	Check the seed laser bandwidth and center wavelength.
Pump laser power is low or unstable	See troubleshooting guide in the pump laser user's manual.
Stretcher misalignment is broadening pulses	Check the alignment of the stretcher. Verify the seed laser beam alignment is optimized.
Pockels cells timing is incorrect	Check the settings for OUT 1 DELAY and OUT 2 DELAY.
Optical components are damaged	Contact your Spectra-Physics representative.
"Wings" are present on output autocorrelation	Check the seed laser bandwidth and center wavelength. Make certain that stretcher mirror M_2 is at focal point of large gold mirror M_1. Verify the compressor grating is parallel to the stretcher grating (coupled grating mount systems only—not applicable to the *Spitfire 50FS*). Contact your Spectra-Physics representative if this is not the case. Verify the beam is not clipping the internal apertures.

Symptom: Output power or output spectrum is unstable

Possible Cause:	Corrective Action:
Power variation in the pump laser	See troubleshooting guide in the pump laser *User's Manual*.
Power variation in the regenerative amplifier	Check the settings for OUT 1 DELAY and OUT 2 DELAY. Check chiller flow and water level.
Spectrum modulated:	
Incorrect adjustment of the ¼ wave voltage to one or both Pockels cells	Contact your Spectra-Physics representative.
Excessive jitter on Spitfire output pulse:	
Unstable seed laser performance	See troubleshooting guide in the seed laser user's manual.
Incorrect timing of the INPUT POCKELS CELL	Check the settings for OUT 1 DELAY.
Defective SDG II or Failure of high speed driver(s)	Contact your Spectra-Physics representative.

Symptom: Poor contrast ratio

Possible Cause:	Corrective Action:
Pre-pulse:	
Incorrect alignment of the output Pockels cell	Contact your Spectra-Physics representative
Incorrect alignment of the ¼ wave plate for the input Pockels cell	Contact your Spectra-Physics representative
Post-pulse:	
Incorrect alignment of the Input Pockels cell	Contact your Spectra-Physics representative
Incorrect adjustment of the ¼ wave voltage for Input Pockels cell	Contact your Spectra-Physics representative

Symptom: Poor output beam quality

Possible Cause:	Corrective Action:
Incorrect pump beam alignment	Refer to Appendix C for pump beam alignment procedures.
Damage to optical components	Check for optical damage; contact your Spectra-Physics representative if present.
Compressor vertical retro-reflector and/or horizontal retro-reflector are incorrectly aligned in the horizontal axis	Contact your Spectra-Physics representative.

Symptom: Optical damage in the amplifier cavity

Possible Cause:	Corrective Action:
Seed laser not well modelocked (CW breakthrough)	Contact your Spectra-Physics representative.
Partial restriction of the stretched spectrum	Contact your Spectra-Physics representative.
Failure to remove alignment tools from optical path after checking stretcher or compressor alignment	Contact your Spectra-Physics representative.
Incorrect alignment of amplifier cavity	Contact your Spectra-Physics representative.

Customer Service

At Spectra-Physics, we take great pride in the reliability of our products. Considerable emphasis has been placed on controlled manufacturing methods and quality control throughout the manufacturing process. Nevertheless, even the finest precision instruments will need occasional service. We feel our instruments have excellent service records compared to competitive products, and we hope to demonstrate, in the long run, that we provide excellent service to our customers in two ways: first by providing the best equipment for the money, and second, by offering service facilities that get your instrument repaired and back to you as soon as possible.

Spectra-Physics maintains major service centers in the United States, Europe, and Japan. Additionally, there are field service offices in major United States cities. When calling for service inside the United States, dial our toll free number: **1 (800) 456-2552**. To phone for service in other countries, refer to the section "Service Centers" on page 8-8.

Order replacement parts directly from Spectra-Physics. For ordering or shipping instructions, or for assistance of any kind, contact your nearest sales office or service center. You will need your instrument model and serial numbers available when you call. Service data or shipping instructions will be promptly supplied.

To order optional items or other system components, or for general sales assistance, dial **1 (800) SPL-LASER** in the United States, or **1 (650) 961-2550** from anywhere else.

Warranty

This warranty supplements the warranty contained in the specific sales order. In the event of a conflict between documents, the terms and conditions of the sales order shall prevail.

Unless otherwise specified, all parts and assemblies manufactured by Spectra-Physics are unconditionally warranted to be free of defects in workmanship and materials for a period of one year following delivery of the equipment to the F.O.B. point.

Liability under this warranty is limited to repairing, replacing or giving credit for the purchase price of any equipment that proves defective during the warranty period, provided prior authorization for such return has been given by an authorized representative of Spectra-Physics. Spectra-Physics will provide at its expense all parts and labor and one-way return shipping of the defective part or instrument (if required). In-warranty repaired or replaced equipment is warranted only for the remaining portion of the original warranty period applicable to the repaired or replaced equipment.

This warranty does not apply to any instrument or component not manufactured by Spectra-Physics. When products manufactured by others are included in Spectra-Physics equipment, the original manufacturer's warranty is extended to Spectra-Physics customers.

When products manufactured by others are used in conjunction with Spectra-Physics equipment, this warranty is extended only to the equipment manufactured by Spectra-Physics.

This warranty also does not apply to equipment or components that, upon inspection by Spectra-Physics, discloses to be defective or unworkable due to abuse, mishandling, misuse, alteration, negligence, improper installation, unauthorized modification, damage in transit, or other causes beyond the control of Spectra-Physics.

This warranty is in lieu of all other warranties, expressed or implied, and does not cover incidental or consequential loss.

The above warranty is valid for units purchased and used in the United States only. Products shipped outside the United States are subject to a warranty surcharge.

Return of the Instrument for Repair

Contact your nearest Spectra-Physics field sales office, service center, or local distributor for shipping instructions or an on-site service appointment. You are responsible for one-way shipment of the defective part or instrument to Spectra-Physics.

We encourage you to use the original packing boxes to secure instruments during shipment. If shipping boxes have been lost or destroyed, we recommend that you order new ones. We will return instruments only in Spectra-Physics containers.

Warning! 

Always drain the cooling water from the laser head and chiller before shipping. Water expands as it freezes and will damage the laser. Even during warm spells or summer months, freezing may occur at high altitudes or in the cargo hold of aircraft. Such damage is excluded from warranty coverage.

Service Centers

Belgium

Telephone: (32) 0800 1 12 57

France

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Spectra-Physics GmbH
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Japan (West)

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(800) 775-5273 (Sales) or
(650) 961-2550 (Operator)
Fax: (650) 964-3584
e-mail: service@splasers.com
sales@splasers.com
Internet: www.spectra-physics.com

**And all European and Middle Eastern countries not included on this list.*
***And all non-European or Middle Eastern countries not included on this list.*

Appendix A RS-232 Interface

Most functions of the *SDG II* can be controlled by any computer with a standard RS-232 serial port. The RS-232 command syntax described here is designed to replicate the functions of the front panel controls and read-outs of the *SDG II* controller.

RS-232 Connector Wiring

The *SDG II* serial port accepts a standard 9-pin D-sub connector male/female extension cable for hookup. Only three pins on the connector are used for serial communications:

Pin Number	Function
2	SDG II transmit data, computer receive data
3	SDG II receive data, computer transmit data
5	Signal ground

RS-232 Communication Protocols

The following protocols must be set in the communication software used to control the *SDG II*:

Setting	Value
Rate	9600 bps
Data Bits	8
Parity	None
Stop Bits	1
Flow Control	None

Command/Query/Response Format

All *SDG II* RS-232 commands, queries, and responses are in ASCII format, and each command or query must be terminated with a carriage return (<CR>). Commands that have a numerical argument must be sent with all the digits, preceded with zeros if necessary.

Commands must all be in lower case. All queries end with a question mark (?). Valid queries return data followed by a carriage return. Valid commands return the string "Ok". Invalid commands or queries return the string "Bad".

Table A-1: Quick Command Reference Guide

Command	Parameter	Function
status?	none	Returns the overall status of the SDG II (see below)
set:cN #	0, 1	Enables (1) or disables (0) the output on channel N (1–3)
read:cN?	none	Returns the output state of channel N (1–3)
set:del:cN ####.#	0000.0 to 1275.0	Sets the delay for channel N (1–3) in nanoseconds
read:del:cN?	none	Returns the delay for channel N (1–3) in nanoseconds
set:rate ####	0001, 0002, etc.	Sets the trigger rate divisor
read:rate?	none	Returns the trigger rate divisor
read:bwd?	none	Returns the state of the BWD latching interlock
reset:bwd	none	Resets the BWD latching interlock
read:sta:bwd?	none	Reads the state of the BWD photodiodes
set:rf #	0, 1	Enables (1) or disables (0) the RF sync
read:rf?	none	Returns the state of the RF sync
set:mode #	0, 1	Sets the trigger mode to continuous (0) or single shot (1)
read:mode?	none	Returns the state of the trigger mode
man:trig	none	Manually triggers the SDG II when in single shot mode

Full Command Description

status?

Returns the status of the *SDG II* as a comma-delimited list of eleven parameters, whose values are shown in the following table:

Parameter	# of Characters	Possible Values
Output 1 state	1	0 (Off) or 1 (On)
Output 2 state	1	0 (Off) or 1 (On)
Sync Out state	1	0 (Off) or 1 (On)
Output 1 Delay	6	0000.0 ns to 1275.0 ns
Output 2 Delay	6	0000.0 ns to 1275.0 ns
Sync Out Delay	6	0000.0 ns to 1275.0 ns
Trigger divisor	4	0001 or 0010
BWD switch state	1	0 (Off) or 1 (On)
BWD photodiode & Ext Interlock state	3	000 to 111 see below under read:sta:bwd?)
Mode	1	0 (continuous) or 1 (single shot)
RF Sync state	1	0 (Off) or 1 (On)

Note

For the following four commands, channel N=1 selects OUT 1 DELAY, channel N=2 selects OUT 2 DELAY, and channel N=3 selects SYNC OUT DELAY.

set:cN #

Sets the output of channel N to be enabled (1) or disabled (0).

read:cN?

Returns the output state of channel N as enabled (1) or disabled (0).

set:del:cN ####.#

Sets the delay of channel N in nanoseconds (ns). The minimum increment for the *SDG II* is 0.25 ns. The allowed values for the last digit (after the decimal) are 0, 2, 5, and 7, which corresponds to 0.00, 0.25, 0.50, and 0.75 ns, respectively. Last digits, other than 0, 2, 5, or 7, are rounded down to the nearest allowed value.

read:del:cN?

Returns the delay setting for channel N. The allowed values for the last digit (after the decimal) are 0, 2, 5, and 7, which corresponds to 0.00, 0.25, 0.50, and 0.75 ns, respectively.

set: rate ####

Sets the divisor by which the input trigger frequency (rep rate) is divided in order to produce the desired output trigger frequency. Allowed values are 0001, 0002, 0005 and 0010. For example, if the input trigger rep rate is 1.000 kHz, a rate of 0005 will set the output frequency to 0.200 kHz.

read: rate?

Returns the input/output frequency divisor set by the set:rate command.

read:bwd?

Returns the state (0=off, 1=on) of the BWD mechanical switch on the back of the *SDG II*.

reset:bwd

Resets the BWD latching interlock. If the BWD switch is on and both BWD photodiodes (PD_1 and PD_2) are illuminated, reset:bwd will clear the BWD latching interlock. If the BWD switch is off, reset:bwd will clear the BWD latching interlock regardless of the state of the BWD photodiodes.

read:sta:bwd?

Returns a string of three binary values. The first two values are the states of the BWD photodiodes (PD_1 and PD_2), where 0=off and 1=on. The third value is the state of the +5 Vdc interlock, where 0=latched and 1=clear). For example, “110” indicates that PD_1 and PD_2 are illuminated but the +5 Vdc interlock is latched, preventing output.

set:rf#

Sets the state of the RF sync to be enabled (1) or disabled (0).

read:rf?

Returns the state of the RF sync as enabled (1) or disabled (0).

set:mode #

Sets the output trigger mode to continuous (0) or single shot (1).

read:mode?

Returns the output trigger mode as continuous (0) or single shot (1).

man:trig

Executes a single output event when the *SDG II* is in single shot mode.

Limitations of RS-232 Control of the SDG II

The following functions cannot be accessed with RS-232 commands:

- The value in the Trigger Frequency display cannot be read.
- The status of the Sync Enable Error LED cannot be read.
- The state set by the BWD on/off mechanical switch cannot be changed.
- The state set by the Interlock enable/disable mechanical switch cannot be changed.

Typical Command Usage

The following scenario illustrates a simple control sequence when using the RS-232 command language with the *SDG II*:

1. Turn on the system, then wait at least 5 seconds for the *SDG II* to initialize.
2. **status?** Determine the state of the *SDG II*.
3. **set:cN** Enable the required outputs.
4. **set:del:cN** Set the required delay values.
5. **set:rate** Set the output trigger frequency.
6. **reset:bwd** If all interlocks are cleared, enable output.
7. **status?** Periodically monitor the *SDG II*.

Appendix B Changing to/from PicoMask Operation

A General Note on Changing Spitfire Versions

It might be possible to change the output characteristics of a *Spitfire* amplifier, but conversion depends upon the amplifier model—not all systems can be converted to other versions. Refer to Chapter 1 for a complete description of the different versions of the *Spitfire* amplifier.

It is also possible, with the proper sets of optics, to extend the wavelength of the output of most versions to a portion of the range between 750 nm and 900 nm. In addition, if ordered from the factory with this option, it is possible to change the output of most amplifiers to either 1 kHz or 5 kHz pulse repetition rate.

While most often straightforward, it is possible that conversion between *Spitfire* models might require alignment techniques that are beyond the scope of this manual. For more information about changing wavelengths or pulse repetition rates, contact Spectra-Physics.

Converting between PicoMask and Femtosecond Operation

Spitfire PM systems are assembled and tested at the factory so that they can be transformed in the field to either a *Spitfire F* (<130 fs pulse width) or *Spitfire USF* (<90 fs pulse width). Similarly, if ordered with this option, a *Spitfire F* or *Spitfire USF* amplifier can be converted to a *Spitfire PM*, which can produce picosecond pulses (2 ps pulse width) when seeded by a femtosecond *Mai Tai* or *Tsunami* laser.

The necessary parts for conversion are included with each system. This appendix lists the procedures for changing between these versions of the *Spitfire* amplifier.

Tools Required

- hex driver for M3 (for some versions of *Spitfire*)
- hex driver for ¼–20 screws
- hex driver for 0.050 in. screws
- $^{3}/_{16}$ in. hex driver
- IR viewer

Changing the Spitfire PM to Femtosecond Operation

Warning!

Do not attempt to clean the surfaces of the gratings or the gold-coated mirrors! These optical surfaces can only be blown clean with dry nitrogen. Attempting to clean these components will permanently damage them! Do not allow *anything* to touch their surfaces!

1. Block the seed and pump beams or close the shutter on these lasers.
2. Using the $^{3}/_{16}$ in. hex driver, remove the mask assembly (Figure B-1) that is in front of the gold mirror M_1 in the stretcher (Figure B-2 and Figure B-3). Do not loosen or move the block used to position the mask assembly; leave it in place in order to return the mask assembly to its correct position when this procedure is reversed.

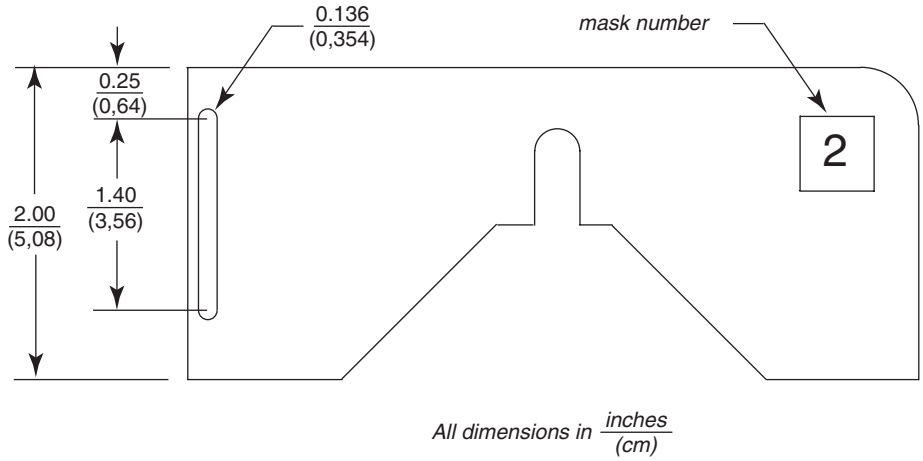


Figure B-1: Stretcher Mask

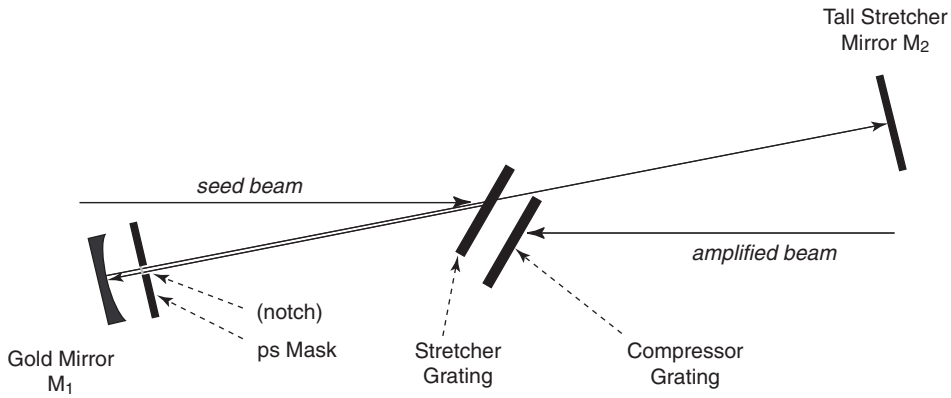


Figure B-2: Modifications to the Stretcher for PicoMask Operation

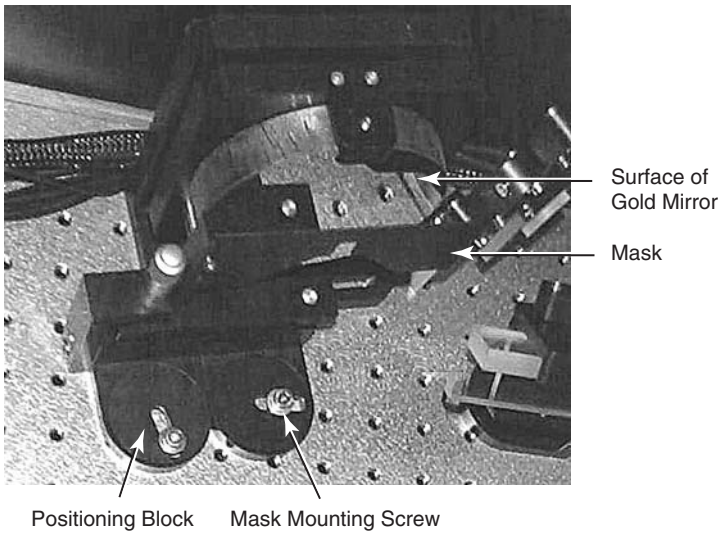


Figure B-3: PicoMask Assembly Mounting

3. Remove the picosecond grating assembly from the rotation stage by removing the two grating mounting screws (they are either ¼–20 or M3 screws—see Figure B-4). Store the grating assembly carefully. Often the assembly can be stored in the stretcher compartment by bolting it to the base plate next to the wall by the Tall Stretcher Mirror.

Note

The configuration of the mask mount differs depending on the date of manufacture of the *Spitfire*.

Warning!

Take care that the Allen wrench or hex driver does not touch the surface of the grating, which is close to the mounting screws.

4. Loosen the two ¼–20 screws that secure the rotation stage, and slide the stage forward to the femtosecond position as marked on the base plate of the amplifier assembly (see Figure B-4).
5. Tighten the two ¼–20 screws to secure the rotation stage in the femtosecond position.
6. Place the *Spitfire F* or the *Spitfire USF* femtosecond grating assembly on the rotation stage, and secure it using the two mounting screws that were removed in Step 3 (see Figure B-5).
7. Unblock or unshutter the seed laser to allow the seed beam to enter the stretcher.
8. Using the IR viewer, rotate the grating stage until the correct femtosecond pattern on the stretcher grating is observed.

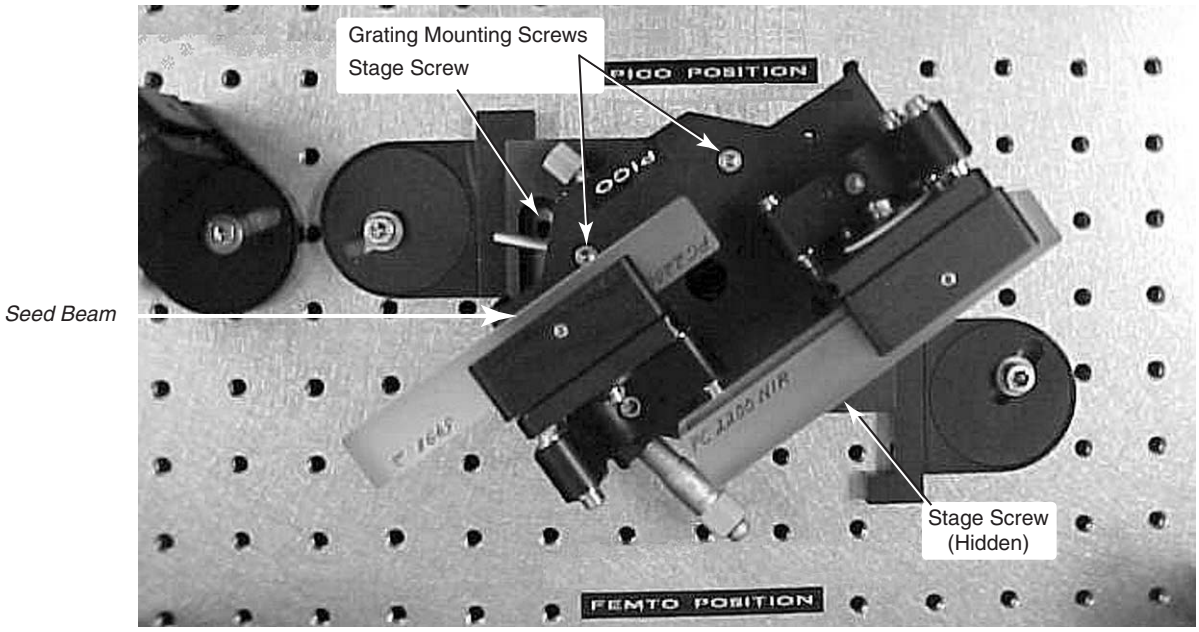


Figure B-4: Rotation Stage, Picosecond Configuration

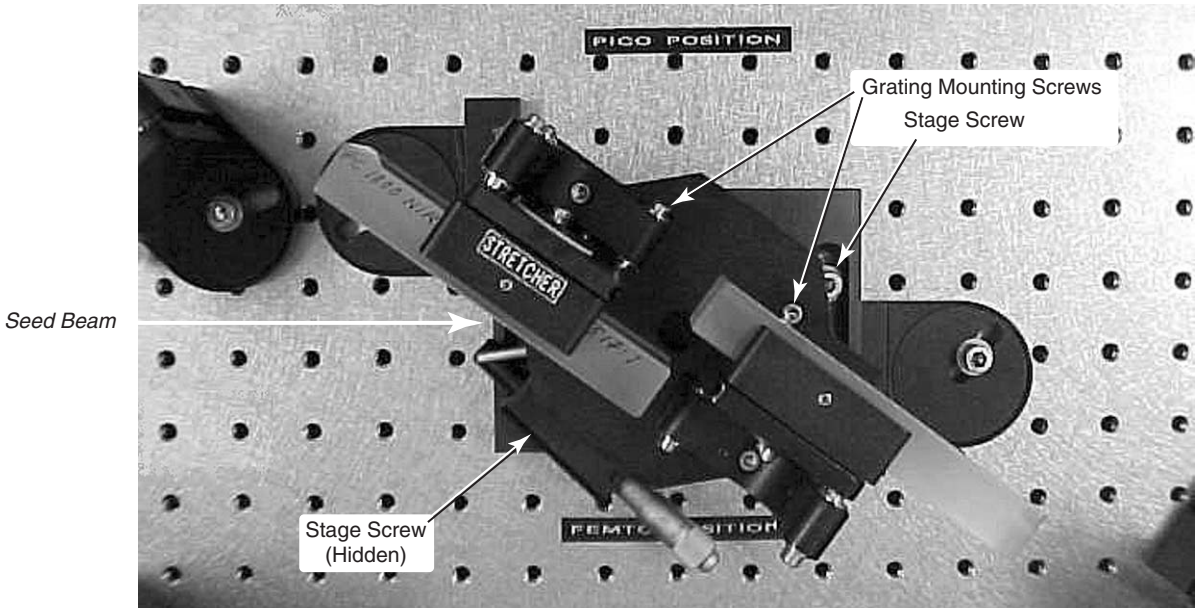


Figure B-5: Rotation Stage, Femtosecond Configuration

9. Next, adjust the BWD photodetectors for the new pulse bandwidth in the stretcher while observing the signals for PD_1 and PD_2 on the *SGD II*.

   a. Loosen the 0.050 in. setscrews on top of the BWD photodetector slide assembly behind the tall mirror, M_2 (Figure B-6). For femtosecond operation, move the photodiodes apart until the LEDs on the *SGD II* just flicker, then slide them together slightly until they produce a bright and steady glow.

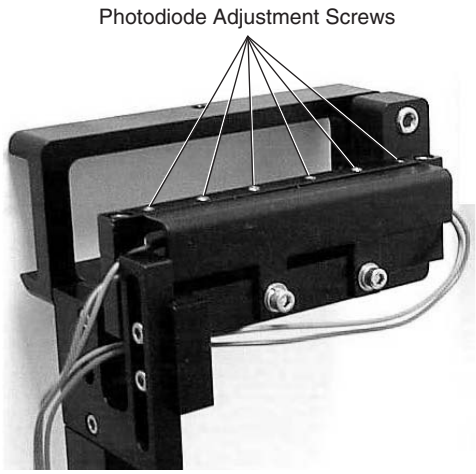


Figure B-6: Adjustment Screws for the BWD Photodiodes

Note

The design of the adjustment for the BWD photodiodes differs depending on the date of manufacture of the *Spitfire*.

b. If either or both BWD LEDs are not brightly lit, then *slightly* adjust gold mirror M_1 up or down until both LEDs produce a bright and constant display.

c. Compensate for any adjustment of M1 with the opposite adjustment of the tall mirror M_2 so that the fourth pass of the beam in the stretcher is picked off by M_3. Refer to the instructions in Chapter 6 for aligning the seed beam into the regenerative amplifier.

10. Use the IR viewer to check the pattern on the compressor grating. If necessary, translate the compressor stage to obtain the correct pattern.
11. The *Spitfire* should now be ready for femtosecond operation.

Converting the Spitfire F to PicoMask Operation

A *Spitfire F* or a *Spitfire USF* may be converted to a *Spitfire PM* system if the system has been configured and tested at the factory for this option.

This procedure is very similar to the inverse procedure, that is, converting a *Spitfire PM* system to one of the femtosecond amplifiers. Refer as needed to the figures used in the inverse procedure described in the previous section.

1. Block the seed and pump beams or close the shutter on these lasers.
2. Using the $^3/_{16}$ in. hex driver, place the mask assembly on the mount in front of gold mirror M_1 in the stretcher (see Figure B-3).
3. Remove the femtosecond grating assembly from the rotation stage by removing the two grating mounting screws (they are either ¼–20 or M3 screws—see Figure B-4). Store the PicoMask grating assembly carefully. Often the assembly can be stored in the stretcher compartment by bolting it to the base plate next to the wall by the tall stretcher mirror.

Note 

The configuration of the mask mount differs depending on the date of manufacture of the *Spitfire*.

Warning! 

Take care that the Allen wrench or hex driver doesn't touch the grating surface, which is close to the mounting screws.

4. Loosen the two ¼–20 screws that secure the rotation stage, and slide the stage backward to the picosecond position as marked on the base plate of the amplifier assembly (see Figure B-4).
5. Tighten the two ¼–20 screws to secure the rotation stage in the picosecond position.
6. Place the *Spitfire PM* picosecond grating assembly on the rotation stage and secure it using the two mounting screws removed in Step 3.
7. Unblock or unshutter the seed laser and allow the seed beam to enter stretcher.
8. Check the alignment of the beam through the mask. The beam reflects from mirror M_1 multiple times. Make sure that only the top beam is clipped by the top (narrow) notch of the mask. Be certain that the spectrum reflected back from M_1 is centered on the notch of the mask.
9. Using the IR viewer, rotate the grating stage until the correct picosecond pattern on the stretcher grating is observed.
10. Next, adjust the BWD photodetectors for the new pulse bandwidth in the stretcher while observing the signals for PD_1 and PD_2 on the *SGD II*:
    a. Loosen the 0.050 in. setscrews on top of the photodetector slide assembly behind the tall mirror, M_2 (see Figure B-6). For picosecond operation, move the photodiodes closer together until the LEDs on the *SDG II* just flicker, then slide them apart slightly until they produce a bright and steady glow.
    b. If either or both BWD LEDs are not brightly lit, then *slightly* adjust gold mirror M_1 up or down until both LEDs produce a bright and constant display.
    c. Compensate for any adjustment of M_1 with the opposite adjustment of the tall mirror M_2 so that the fourth pass of the beam in the stretcher is picked off by M_3. Refer to the instructions in Chapter 6 for aligning the seed beam into the regenerative amplifier.
11. Use the IR viewer to check the pattern on the compressor grating. Make sure the first and the last beam spots on the compressor grating are in a single vertical line. If necessary, translate the compressor stage to obtain the correct pattern.
12. The *Spitfire* should now be ready for picomask operation.

Appendix C Alignment

Before starting these procedures, it is essential that you read Chapter 2, "Laser Safety," and that you become thoroughly familiar with the components and optical design of the *Spitfire* as discussed in Chapter 7.



Use of controls or adjustments, or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Caution! 

The following procedures are not intended for the initial installation of the *Spitfire* amplifier. Call your Spectra-Physics service representative to arrange an installation appointment, which is part of your purchase agreement. *Allow only authorized Spectra-Physics representatives to install your laser.* You will be charged for repair of any damage incurred if you attempt to install the system yourself, and such action may void your warranty.

These procedures are supplied as a convenience in the event your *Spitfire* system is out of warranty and a service call is problematic. These advanced procedures might well result in loss of function or even permanent damage to the system if performed by personnel not trained by Spectra-Physics.

If the amplifier is no longer lasing, or if an intracavity optical component is seriously misaligned or damaged and must be replaced, contact your Spectra-Physics representative before attempting any repair. Experienced experts may be able to apprise you of techniques that might save you considerable time and expense in these circumstances.

More advanced procedures, such as replacing a damaged Ti:sapphire rod in the amplifier, will require a service call.

Essential to proper *Spitfire* amplifier operation is the alignment of the seed laser into the *Spitfire*, amplifier optimization and other similar adjustments. These are considered routine and are described in Chapter 6, "Operation."

If the amplifier is operating properly with the installed optics set, but operation at a different wavelength range is required, contact your Spectra-Physics representative.

If converting the *Spitfire* from femtosecond (*Spitfire F* or *Spitfire USF*) to picosecond operation (*Spitfire PM*), or the reverse, refer to the relevant procedures in Appendix B. It is not necessary to realign the amplifier cavity for these procedures.

To understand these procedures, it is important to realize the *Spitfire* can operate as a laser as well as an amplifier—that is, it can be configured to produce its own pulsed output (when energized by the pump laser) even when the input from the seed laser is blocked.

Begin these procedures with the pump laser and the seed laser on and warmed up, with both beams blocked (shuttered) from entering the *Spitfire*.

Try This First

If your system is lasing but performance has degraded, slight adjustments might only be required in pump beam power or to the pump beam routing mirror, PM_2 (refer to Figure C-3) to optimize output, rather than performing a complete realignment.

Before beginning any realignment, verify the following:

- there is sufficient pump power
- the pump beam has not been misaligned
- the *SDG II* OUT 2 DELAY is sufficiently beyond OUT 1 DELAY (see "Basic Performance Optimization" on page 6-3.
- the +5VDC ENABLE switch on the *SDG II* back panel is in the disabled or down position.
- the Pockels cells are properly connected, triggered, and operating
- the intracavity apertures are not blocking the beam

Tools Required:

- a power meter capable of measuring between 10 mW and 20 W of average power from 500 nm to 900 nm
- a fast CRT analog oscilloscope capable of 300 MHz or better
- a fast photodiode with a 2 ns rise time or better
- IR viewer and IR card
- a small, low-divergence HeNe laser (for alignment)
- an autocorrelator (e.g., Spectra-Physics Model SSA)
- scales (rulers) to measure up to 10 in. and 25 cm
- three gimbal mounts with 4–6 in. adjustable height
- three silver mirrors for the above mounts
- alignment pins
- #1 Phillips screwdriver
- a standard (English) hex driver set
- a standard (English) hex ball driver set
- a metric hex driver set
- white business card
- trim pot screwdriver
- lens tissue
- gel linear polarizing film

Stretcher Alignment Check

If you cannot see the beam in the amplifier, it will be necessary to check the alignment through the stretcher, as follows:

1. Use an IR viewer to look at the beam pattern on the stretcher grating. It should look like that in Figure C-1.

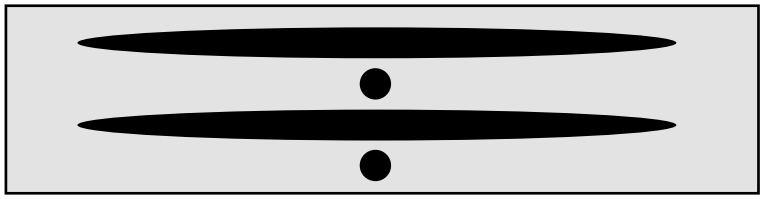


Figure C-1: Radiation Patterns on Stretcher Gratings

2. If the beam pattern does not look like Figure C-1, then it is likely that the wavelength of the seed laser has changed. In order to return to the previous operating conditions, adjust the seed laser wavelength until the pattern is symmetrical on the grating as shown here.

   After adjusting the seed laser, re-check the beam pattern. If the pattern shown in Figure C-1 cannot be obtained, the stretcher may need to be realigned. This procedure is beyond the scope of this manual. Contact Spectra-Physics for assistance.

3. The output beam from the stretcher should now be picked off by mirror M_3. It may be necessary to *slightly* adjust the vertical tilt of the large gold mirror M_1 using the adjustment described in Chapter 7, "Stretcher and Compressor Beam Paths."
4. Re-check the alignment of the beam into the regenerative amplifier.

Compressor Alignment Check

The alignment of the beam through the compressor should not have changed since the *Spitfire* was last operated, but check it anyway.

Verify the output beam is round and even in intensity.

Use an IR viewer to look at the compressor grating. The pattern shown in Figure C-2 should be evident.



Figure C-2: Radiation Patterns on Compressor Gratings

Pump Beam Alignment

Once the *Spitfire* system has been properly installed, the alignment of the pump laser into the *Spitfire* amplifier should not need to be adjusted during normal operation. Some circumstances however, such as maintenance or service of the pump laser, may require aligning the pump beam. The most common symptom of pump beam misalignment is poor *Spitfire* mode quality that results when there is poor superposition of the pump beam mode in the Ti:sapphire rod.

Refer to Chapter 7 for a description of the optical design shown in Figure C-3.

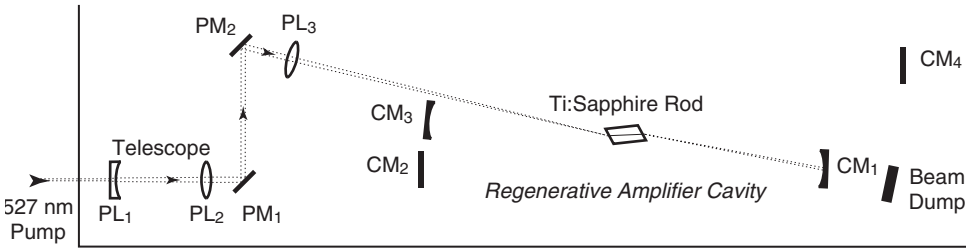


Figure C-3: Pump Beam Path of the *Spitfire*

1. Record the pump beam power required to operate the *Spitfire*.
2. Block the seed pulses into the *Spitfire*.
3. Adjust the pump laser (*Evolution*) power to the minimum power that allows a stable green beam to be observed.
4. Verify the pump beam passes through the input port without clipping.
5. Adjust the pump beam to center it on the lenses of the zoom telescope (PL_1 and PL_2) and also on PM_1.
6. Adjust PM_1 to center the beam on PM_2.
7. Adjust PM_2 to center the beam in the Ti:sapphire rod.
8. The beam should be 1 to 2 mm from the edge of mirror CM_3. If this is not the case, make small adjustments to PM_1 as needed.
9. When the pump beam is centered on the Ti:sapphire rod, it should pass through the center of PL_3. If it does not, loosen the screw holding the mount for PL_3 to the chassis and position it so that it does. Be sure to maintain the distance from the lens to the face of the Ti:sapphire rod. If moving PL_3 moves the pump beam on the rod, use PL_2 to re-center it. Iterate adjustment of PL_2 and PL_3 until the beam is centered on both PL_3 and the rod.

   Note that having the correct distance from PL_3 to the Ti:sapphire rod is particularly important for maintaining proper pump beam mode in the rod. The correct placement of PL_3 should be marked on the base plate of the amplifier head assembly.
10. Return the pump laser to Q-switched operation, and adjust it to the power level noted in Step 1. If the required pump power is not known, set the pump power to 10 W.

11. The *Spitfire* should now operate as a laser. If it does not, scan the pump beam waist across the Ti:sapphire crystal face until it is superposed onto the intracavity beam waist. To do this, position the pump beam rather high in the crystal and off to one side using PM_2. Now slowly translate the pump beam across the crystal face to the opposite side. Lower the beam position in the crystal ~0.5 mm and slowly translate back across the crystal face. Continue this scanning process until the amplifier resonator begins to lase.

Note



The Ti:sapphire crystal emits *less* fluorescence when the *Spitfire* begins to lase. While scanning, watch for this, rather than for an output from the thin-film polarizer, which necessitates watching *both* the crystal (for safety) *and* the output (for lasing).

12. Once the amplifier has begun to lase, position a power meter just before the second lens of the beam expanding telescope, OL_2.
13. Adjust PM_2 vertically and horizontally for a symmetrically shaped output mode (it should approach a single-order mode). This should also coincide with maximum output power.
14. Verify the cavity beams are parallel to the chassis top surface (measure the mirror *leakage* beam height from the chassis surface *outside* the cavity behind CM_3 and CM_4) and correct any error.

    It is likely that the beam height beyond CM_4 is either too high or too low. To correct any error, make vertical adjustments to CM_3 while compensating for power and mode shape with vertical adjustments to CM_4 until the correct beam height is restored.



Laser radiation is present. Beware the eye hazard from the residual pump beam behind CM_2!

15. Repeat Step 14 for the beam between CM_4 and CM_1. Adjust CM_4 for beam height behind CM_1 and compensate for output power and mode shape with CM_1.
16. Reposition both intracavity apertures coaxially about the beam as follows. Open each aperture fully, then reduce the diameter of each iris. The beam should be symmetrical around the iris as it is reduced, passing through its center. If it is not, loosen the screw that retains the aperture post and center the iris about the 800 nm intracavity beam.
17. Open both intracavity apertures, and record the output power.

The pump laser beam should now be optimally aligned.

Compressor Alignment

Danger!
Laser Radiation

In the following procedure, the regenerative amplifier is initially operated as an optically pumped Q-switched laser. In this configuration, the *Spitfire* is capable of producing >1.5 W of average power at a wavelength near 800 nm. Use appropriate caution.

This procedure assumes that the grating block assembly is properly aligned, as are the input beams into the stretcher and compressor, and that the *Spitfire* is fully operational. Under these circumstances, the compressor requires only minimal adjustment for optimal performance.

1. Disable OUT 1 DELAY and OUT 2 DELAY on the *SDG II*.
2. Close the shutters of the pump laser and the seed laser.

Warning!

Failure to block the seed beam when called for in this procedure will result in significant damage to amplifier components. Such damage is not covered by your warranty.

3. Remove the covers from the *Spitfire*.
4. **Carefully** remove the grating block by removing the ¼–20 or M3 screws (depending on the *Spitfire* revision) from the rotation stage.

   If aligning a *Spitfire 50FS* compressor, remove only the compressor grating.
5. Install the removable reference iris in the X_1 location (see Figure C-4), and adjust the aperture opening to about 4 mm diameter.

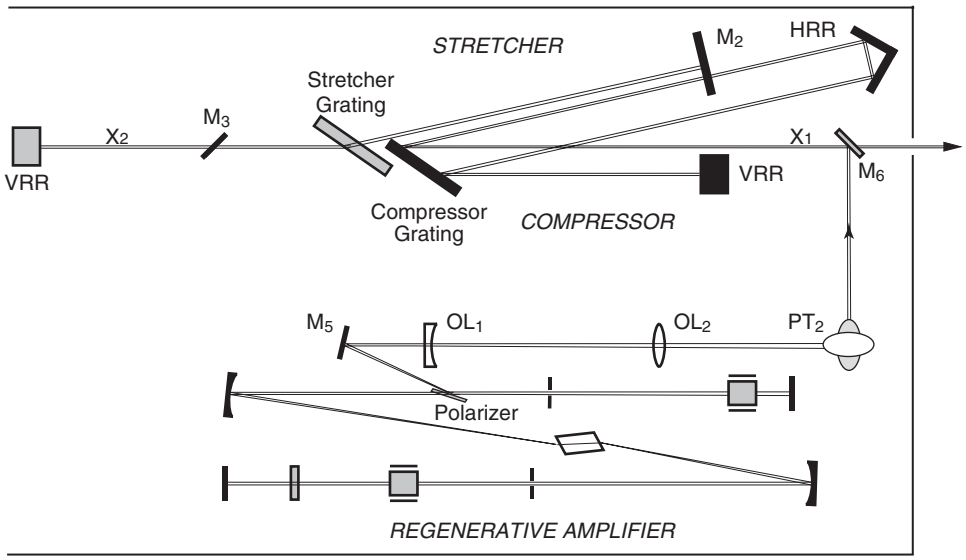


Figure C-4: Alignment of beam into the compressor

6. Operate the *Spitfire* as a Q-switched, cavity-dumped laser. To do this, enable OUT 1 DELAY and OUT 2 DELAY and adjust OUT 2 DELAY as necessary.

   Remember that the seed beam is currently blocked from entering the *Spitfire*.
7. Adjust mirror M_5 to center the cavity-dumped beam through the apertures of the removable iris.
8. Relocate the removable iris to location X_2, as shown in Figure C-4.
9. Adjust mirror M_6 to center the cavity-dumped beam through the apertures of the removable iris.
10. Disable OUT 1 DELAY and OUT 2 DELAY on the *SDG II*.
11. Close the pump beam shutter.
12. Install the grating assembly (or the compressor grating assembly for the *Spitfire 50FS*).
13. Open the seed beam input shutter.
14. Adjust the rotation of the grating assembly so that the pattern on the stretcher grating appears as shown in Figure C-1.
15. Close the seed beam input shutter.
16. Open the pump beam shutter.
17. Enable OUT 1 DELAY and disable OUT 2 DELAY trigger pulses.
18. Open the seed beam input shutter. Verify the buildup time reduction: it should be the same as when the seed beam optimization alignment procedure is performed.
19. Enable the OUT 2 DELAY trigger on the *SDG II* and adjust the timing for cavity dumping the correct pulse.
20. Verify that the cavity-dumped output is centered on the iris.
21. Using the IR viewer to look at the grating, verify the pattern on the grating appears the same as in Figure C-2. If not, go back and check the alignment through the iris.
22. Remove the iris from the *Spitfire*, then verify the output beam is not clipped.

Warning! 

Failure to remove the iris when called for in this procedure will result in significant damage to *Spitfire* components. Such damage is not covered by your warranty.

23. The compressor alignment should now be optimized. If this is not the case, contact your Spectra-Physics service engineer.

Ejecting the Pulse from the Amplifier

After performing the alignment procedures above, it will be necessary to adjust the timing of the Pockels cells to achieve proper capture and ejection of pulses from the amplifier. Refer to Chapter 6 for this procedure.

Notes

Report Form for Problems and Solutions

We have provided this form to encourage you to tell us about any difficulties you have experienced in using your Spectra-Physics instrument or its manual—problems that did not require a formal call or letter to our service department, but that you feel should be remedied. We are always interested in improving our products and manuals, and we appreciate all suggestions.

Thank you.

From:

Name _____

Company or Institution _____

Department _____

Address _____

_____

Instrument Model Number _____ Serial Number _____

Problem: _____

_____

_____

_____

_____

Suggested Solution(s): _____

_____

_____

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